

Stratocracy - Prototype GDD

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Hex turn-based strategy · Unreal Engine 5.8 · 7-week AI-agent jam

1. Executive Summary

Concept. Two commanders battle across a small skirmish-sized hex map. Units move over terrain that shapes movement and defense; players capture factories to reinforce and win by destroying the enemy's **flag unit**. Matches run **~10–15 minutes typical**, with the turn cap (§2.8) putting the **outer bound near 20 minutes** at a normal move pace against the AI — so 10–15 usual, ~20 worst-case if a match goes the distance to the cap. That's an expected duration, not a wall-clock guarantee (a PvP hotseat, a stretch feature, would add a per-turn timer). The pacing discipline is deliberate, see Lineage.

The deliverable is the game; AI agents are the method. The graded, shipped output is a complete, playable prototype — deliberately systems-forward and art-light so that AI agents can author, test, and tune the bulk of it, with the human acting as director. The genre was chosen for exactly this reason: a deterministic hex wargame is almost entirely **rules + data** — the surface agents author most reliably and can self-verify with tests. How much the agents actually did is reported honestly as a per-system provenance ledger (§3), as supporting evidence of the method — but the thing that ships and is judged is the game. (Accepted tradeoff: the art-light choice leaves visual-craft strengths on the bench, by design.)

Lineage. *Conflict* (Vic Tokai, NES, 1989) — the first NES game to render a hex battlefield. We inherit its structure (hex map, terrain modifiers, capturable cities/factories, destroy-the-flag win) and explicitly fix the flaw its own reviewers named: **Famicom Tsūshin (Famitsu) faulted it for battles that “took too long”** (Western outlets were kinder — Nintendo Power 4/5 — so the criticism is specifically about pacing, not the design). Everything here is tuned toward short, decisive play.

Design pillars. 1. **Legible, deterministic rules** — agent-authorable and unit-testable; no hidden RNG (seeded if any). 2. **Short and decisive** — small maps, punchy damage, a turn cap. Never an hour-long slog. 3. **Terrain + unit rock-paper-scissors** — the only depth needed; it comes from data, not feature count. 4. **Minimal art, maximal system.** 5. **Tests are a deliverable** — every core system ships with agent-generated tests and self-play balance data.

Scope at a glance. One polished scenario, four units, six terrains + the capturable Factory tile, capture + production, a heuristic AI opponent, and a functional UI is a *complete, shippable game*. Everything beyond that is garnish (see §2 scope table). Core playable in weeks 1–3; the remaining four weeks are content, balance, UI polish, and documenting the pipeline.

Success = a shipped, playable prototype. The primary bar is the game itself: the §2 core loop running, tuned, and fun in a 10–15 minute match. Agent contribution is tracked as a **per-system provenance ledger** (§3) — an honest, row-by-row record of which systems agents authored and verified — reported as supporting evidence of the method, *not* as the pass/fail bar. Anything a human hand-authored (UI polish, glue) is logged as human, so the ledger under-claims rather than over-claims.

Design decisions (all resolved). 1. ~~Scenarios in the prototype: one polished, or three?~~ → **RESOLVED: one polished scenario.** Content is the most agent-scalable axis; one hand-built map proves the pipeline without spending human review budget on volume. 2. ~~Economy: production only, or add the “fame” scoring spine?~~ → **RESOLVED (updated): a unified Fame currency.** Production points, combat rewards, and the win-score collapse into one **Fame** pool inherited from *Conflict* — earned from held factories and kills, spent to build, tallied to decide a capped match (§2.7). This is now core (not a separate stretch spine); flag-kill remains the primary, decisive win. 3. ~~Opponent: heuristic-only, or attempt the LLM commander?~~ → **RESOLVED: heuristic-only ships (§2.9); LLM commander stays a stretch toggle.** 4. ~~2-player hotseat: in scope or cut?~~ → **RESOLVED: stretch only, off the critical path (§2.10).** Nearly free on IGOUGO, but doesn't serve the

agent-authorship thesis, so it never blocks core. 5. ~~Working title:~~ → **RESOLVED: the game is titled *Stratocracy*.**

1.5 Revision Notes — First Draft → Final Draft

No external feedback was returned on the First Draft in time for this revision, so rather than wait, I **stress-tested the draft with an agent review crew**— the exact technique from the course’s GDD-anatomy stress-test method (an Exploit-Hunter / Consistency / Pacing board run over the document, then human-adjudicated). Applying the course’s own agent method to my own GDD *is* the growth here: the crew surfaced concrete, exploitable logic gaps on paper, and every one below was fixed before a line of code. Changes are listed as **finding** → **change** → **why it’s better**.

#	Finding (agent review crew)	Change	Why it's better
1	Turtle exploit in the tiebreak. Original cap tiebreak (“most factories held, then total HP”) <i>rewarded</i> stalling: grab one factory, wall the flag in a corner, run out the clock, win without fighting.	Reordered the cap tiebreak around combat Fame earned (damage dealt), excluding passive factory income; added a mutual-passivity → draw guard; put multiple factories on the map and a live standings scoreboard (§2.7–2.8, §2.11).	Turtling now <i>loses</i> the cap. Pillar 2 (“short & decisive”) is enforced by the grading itself, not just by the turn cap.
2	Success metric was unmeasurable. “Agents did X% of the work” was a hero-number that couldn’t be checked or defended.	Replaced it with a per-system provenance ledger — one row per system, marked by author and agent-verified status (§3).	It’s <i>checkable</i> (re-runnable tests + commit trace) and deliberately under-claims (human work logged as human).
3	Economy was unpriced. Production points, combat rewards, and win-score were three separate uncapped pools; unit costs were undefined; built units had “nowhere to go / uncapped points.”	Collapsed everything into one Fame currency with explicit costs (Inf 100 / Recon 150 / Arty 200 / Tank 300), factory +100/turn, kill ≈ ½ cost, flag +500 ends the match, and space-throttled spawning (§2.7).	One legible pool the player and the AI both reason about; the uncapped-points gap is closed by throttling on board space, not an arbitrary ceiling.
4	The AI ignored the economy. A re-review found the baseline opponent never built or captured — it would fight with only its starting force, making the whole economy one-sided.	Gave the baseline AI a two-phase turn: an economy (build) phase plus a capture step before the unit phase (§2.9).	Keeps production, capture, and the “objectives held” tiebreak live on <i>both</i> sides — the economy is actually exercised in play.
5	Scoring could invert the win. Raw-Fame scoring let a long cap match’s accumulated kill-Fame “outscore” an actual flag kill, and a mutual-zero-combat cap had no defined result.	Made victory a set of categorical tiers (Decisive > Marginal > Draw); combat Fame is only the sort key inside the tiebreak, never the grade; mutual-zero-combat = draw (§2.8).	A flag kill is <i>always</i> the best outcome — the win condition can never be out-piled by attrition points.
6	Framing risk (design judgment). Positioning the <i>agent method</i> as the deliverable would have misaligned with a game-first rubric that grades the playable product first.	Reframed the identity: the game is the deliverable; agent-driven development is the <i>method</i>, reported as supporting evidence via the ledger (§1, §3).	Aligns the document with how the work is actually judged — the shippable game leads, the pipeline substantiates it.

1.6 Revision Notes — Final Draft → Production Draft

The Final Draft closed the *logic* gaps §1.5 found, but it left the document thin in four specific places that external feedback and the review board had already named. This revision closes them, and it was authored the same way the game is being built: a **four-author crew** — rules, UX/onboarding, scenario, and technical — writing in parallel against a frozen snapshot of this document, each confined to its own section and forbidden to edit the master. A separate **continuity gate** audited every draft against the live GDD for contradictions, stat drift, dead references, and invented numbers, and merge was blocked until it returned PASS. It did not pass on the first attempt: the gate filed four violations (a production-menu mock that greyed a unit the player could afford, a scoreboard chevron marking the wrong leader, and two dead references in the technical draft), the two authors responsible were re-spawned with those violations, and only the corrected drafts merged. That failure-and-correction loop is the same one §3 documents for T-COMBAT-07 — applied to prose instead of code. Changes are listed as **finding** → **change** → **why it's better**.

#	Finding (source)	Change	Why it's better
1	No stated player-experience goals (<i>rubric feedback, -0.5</i>). The document said what the design <i>valued</i> (§1's pillars) but never what the player should <i>feel</i> , so nothing downstream could be checked against an intent.	Added §2.0, six goals PX-1..6, each carrying a rule source and an observable check — e.g. PX-4 “fighting is always better than hiding” is checked by T-CAP-02/03, “a side with zero combat Fame never wins a capped match.”	A goal that cannot be checked is a preference, not a goal. Self-play reports and playtest notes now cite goals by ID, so intent is testable rather than asserted.
2	No onboarding plan (<i>review board open item</i>). The GDD assumed a player who already understood hexes, Fame, capture, and the RPS triangle — with no manual and no tutorial mode in scope, nothing explained how they would learn them.	Added §2.11.6: three teachers (constraint, the forecast, and one-shot event tips), a scripted four-turn guided opening (turns 1–4) delivered through a directive strip, ten one-shot strings plus the two cap-approach banners, a pre-match briefing overlay, and a twelve-row concept ledger mapping every concept to where it is first taught.	The one genuinely unowned risk in a game with no manual is now a specified, buildable surface with a named cut line (§2.11.8) rather than an assumption.
3	The ~3-match replay cliff (<i>review board open item</i>). The game is deterministic end-to-end, so any line that beats the AI once beats it forever; one mirrored map is a solved puzzle by match ~4.	Added §2.13 in full — the shipped scenario <i>Ferrum Crossing</i> specified hex by hex, plus §2.13.4, which reframes the replay unit as a configuration = map × seat × difficulty handicap rather than as new mechanics.	Moves the cliff from match ~3 to match ~6 on shipped scope alone, using scenario data and one menu affordance — no new systems, no new rules, and the stretch maps stay explicitly cuttable.
4	Eight *pending* ledger rows had no route to a gate (§3 <i>provenance ledger</i>). Combat cleared a real test gate; the	Added §4.7's eight gateable spec stubs (Inputs / Formula / Invariants / Determinism / Acceptance, the shape that certified Combat)	Every pending row now has a named acceptance suite and a dependency position, so the ledger flips on evidence rather than on assertion — and

#	Finding (source)	Change	Why it's better
5	other eight systems had no spec, no test IDs, and therefore no way to flip the same way. §2 had drifted after the Conflict fold (<i>continuity audit</i>). Bridge and Factory had been folded into §2.3's terrain set, but §1 and §2.10 still said “five terrains,” and §2.8's tiebreak had grown an apparatus nobody had re-justified.	§4.8 §4.11's build order with an explicit critical path. Consolidated §2.1–§2.10, reconciled the terrain count across §1/§2.3/§2.10, and subjected the whole Fame/tiebreak apparatus to a delete-test — every piece removed in turn to see what breaks (§2.8's closing block).	missing, it is filed as a numbered open question in §4.7's register instead of being invented. The tiebreak now survives on evidence rather than inertia: the one piece that failed its delete-test (the floated per-turn Fame decay) was cut, and what remains is a guard, a sort, and an enum over state the game already tracks.

2. Game Mechanics

2.0 Player-experience goals

The pillars (§1) state what the design values; these goals state what the player should experience, phrased so each is observable in the test suite or the balance harness (§4.1). A goal that cannot be checked is a preference, not a goal — so each row carries its check. Self-play reports and playtest notes cite goals by ID.

ID	The player experiences	Rule source	Observable check
PX-1	Matches feel short and decisive: ~10–15 minutes typical, ~20 worst-case at the cap. A match never becomes the hour-long slog the lineage was criticized for.	§1, §2.8	Self-play turn-length distribution: the median match ends before the turn cap; cap-tiebreak matches are a minority of results.
PX-2	The rules can be trusted completely. The forecast shown before committing an attack is exactly what resolves — a whole turn can be planned in advance.	§2.6	Every attack's forecast equals its resolution; identical inputs give identical results (determinism invariants, T-COMBAT suite).
PX-3	Depth comes from where units stand, not from a memorized counter chart. The triangle is read off movement and range on the board.	§2.4	The type-effectiveness table ships all-1.0; no counter-multiplier is load-bearing at ship.
PX-4	Fighting is always better than hiding. There is no line of play where sealing a corner and running the clock wins.	§2.7, §2.8	A side with zero combat Fame never wins a capped match (T-CAP-02, T-CAP-03).
PX-5	The player always knows who is currently winning and how close the cap is. The tiebreak is never a hidden win condition.	§2.8, §2.11	The standings scoreboard displays every tiebreak input (combat Fame, objectives held X/N , surviving HP) plus the turn counter against the cap.
PX-6	The economy is one thought: earn Fame, spend Fame. The player never converts between currencies or tracks parallel pools.	§2.7	Every income, build cost, and combat reward mutates a single per-side Fame pool; no second resource exists in the data schema.

2.1 Core loop

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Player turn:
  for each of your units (any order):
    select → move (within range, terrain-costed) → act (attack / capture
/ build) → done
  end turn
Opponent turn: same, driven by AI
repeat until a flag unit dies, an objective is met, or the turn cap
triggers a tiebreak

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I-GO-U-GO alternation. No simultaneous resolution.

2.2 Hex grid

The battlefield is a field of pointy-top hexagons. The player reads adjacency and distance at a glance — every hex has six equal neighbours, so no direction is unfairly cheap the way diagonals are on a square grid — and each unit occupies exactly one hex. Line-of-sight blocking (a unit can't see or fire through a ridge) is a stretch goal.

2.3 Terrain (prototype set: 6 movement terrains + the capturable

Factory tile)

Terrain is the first thing the player weighs before moving: some hexes cost more of a unit's movement allowance to enter, some make a unit standing on them harder to kill, and some are closed to certain movement classes entirely — the player is always trading speed against cover and reach. Move cost is per movement class. All values are starter/tuning targets.

Terrain	Move cost	Defense	Passable
Plains	1	0%	land, air
Woods	2	+20%	land, air
Mountains	3	+40%	land (slow), air
Water	—	0%	sea, air
Town	1	+10%	land, air; capturable
Bridge	1	−10%	land, air; the only hex a land unit crosses Water
Factory	1	+15%	land, air; capturable; build/spawn + repair point

Bridge turns Water from dead space into a chokepoint system: land routes must funnel across bridges, and the negative defense makes a unit caught mid-crossing an exposed target — a contested kill-zone, not a safe shortcut. (*Tuning fallback: −10% → 0% if too punishing.*) **Factory** is the economy's anchor (§2.7): occupiable by either side, capturable by Infantry, a defensive anchor, and the build/spawn + repair point.

2.4 Units (prototype set: 4)

Rock-paper-scissors, not a roster. Starter stats.

Unit	HP	Move	Atk	Def	Range	Cost (Fame)	Notes
Infantry	10	3	4	2	1	100	Cheap; the only unit that captures towns/factories
Tank	20	5	8	5	1	300	Line breaker; Flag Unit is a Tank variant (not producible)
Artillery	8	3	10	1	2–3	200	Strong ranged, fragile, no melee counter
Recon/Air	12	7	5	3	1	150	High move; ignores some terrain cost

Cost is in Fame, the single currency (§2.7). All values are tuning targets, scaled down from *Conflict*'s cost ladder for four units and ~20-turn matches (§2.12).

Flag Unit: a designated Tank. Its death = loss. Not producible.

The triangle is positional, not a counter chart. Stratocracy's rock-paper-scissors lives in movement and range, an invariant worth naming: **Artillery beats Tank** (fires from range 2–3, takes no melee counter), **Recon beats Artillery** (move 7 runs it down into a fragile melee), **Tank beats Recon** (higher atk/def wins the range-1 fight). Infantry sits outside the triangle as the capture/objective unit. A type-effectiveness multiplier exists in the combat formula (§3) but ships **defaulted to 1.0 everywhere** — populated only if self-play shows the positional triangle too weak, so depth stays in positioning rather than in a lookup table (Pillar 3; PX-3).

2.5 Movement & pathfinding

The player selects a unit and every hex it can truly reach this turn lights up, terrain costs already accounted for — the highlight is the real move set, not an estimate. Clicking a lit hex sends the unit by the cheapest route. Only one unit fits in a hex, so lanes, chokepoints, and blocking are part of the plan. Zones of control — enemy units freezing anything that

moves next to them — are cut from the prototype; they return only if matches feel too fluid.

2.6 Combat resolution

Before an attack is committed, the game shows the outcome up front: the predicted damage dealt, and the counterattack the defender strikes back with if it survives and the attacker is within its reach (attack forecast, §2.11). Combat is a pure function of attacker, defender, the defender's terrain, and the attacker's remaining health — the same matchup always resolves the same way, so a player can plan a whole turn and trust it plays out as shown (PX-2). Any randomness added later is **seeded**, so a given fight stays reproducible: the forecast the player sees is exactly what resolves, with no hidden roll.

2.7 Economy & capture — the Fame currency

Stratocracy runs on a **single currency, Fame** (inherited from *Conflict*, §2.12): earned from held objectives and combat, spent to build, tallied to decide a capped match. Production points, combat rewards, and the win-score are **one pool, not three** (PX-6). *All numbers are starter/tuning targets, scaled down from Conflict's for four units and ~20-turn matches.*

- **Factories & starting layout:** the map ships with **multiple factories** — a **home factory per side** (owned at start, so both players have income from turn 1) plus **two or more neutral factories** in contested ground; a typical small skirmish map has **~4 factories total**. The spread is what makes expansion worth fighting for and “objectives held” (§2.8) a real 0–N measure instead of a 1–0 coin-flip.
- **Income:** each **factory** held pays **+100 Fame/turn**; each captured **town** pays **+25/turn**. More objectives held = a faster army, so the neutral factories are the mid-game prize.
- **Build & spawn:** spend Fame at a factory to produce a unit from the buildlist (§2.4 costs: Infantry 100, Recon 150, Artillery 200, Tank 300). The unit **spawns on the factory hex if it's free, otherwise an adjacent free hex; if the factory is boxed in, the build waits**. Fame has no hard cap — deployment is throttled by board space, not a point ceiling.
- **Combat Fame:** destroying an enemy unit pays **~half its Fame cost** (e.g. a Tank kill = +150 — see Q5, §4.7, to make this exact); an **undamaged strike** (attacker takes no counter) pays a small bonus (see Q6, §4.7); destroying the enemy **flag pays +500 and ends the match**. These feed the same Fame pool and the cap tiebreak (§2.8).
- **Capture:** move an Infantry (the only capturer) onto a town/factory and hold to capture over N turns (start N=1–2); a captured objective flips its Fame income to the new owner.
- **Starting Fame:** each side opens with **200 Fame** (enough for two Infantry or one Artillery on turn 1) plus home-factory income from turn 1. Single-player difficulty is a starting-Fame handicap — see §2.9.
- **Repair:** a unit that ends its turn on an **owned Town or Factory and is not adjacent to any enemy** heals **+25% of max HP** (rounded down, min 1, capped at max) at the start of its next turn — free for the prototype. This is the game's only HP-recovery path; without it every unit is a one-way asset. The **not-adjacent clause is the anti-fortress lock**: a unit must break contact to repair, so a factory next to the front never becomes unkillable — and because repairing earns zero combat Fame, a repair-turtle still loses the cap tiebreak (§2.8). *(Heal %, and a possible small Fame cost, are tuning levers.)*

2.8 Turn structure & victory

- **Primary win — decisive victory:** destroy the enemy flag unit. Ends the match immediately.
- **Secondary win — territorial domination:** control **every factory on the map** at the start of your turn. Ends the match immediately and is ranked a **Decisive win**, equal to a flag kill. Because taking the last factory means capturing the enemy home factory deep in their territory, a flag kill is usually already available by then — this is an **active backstop that closes out a flag-turtle stalemate before the cap**, not a common win. Factories only (towns excluded), so it stays hard-won.
- **Loss:** your own flag unit is destroyed, or the enemy dominates all factories.

Turn cap → attrition tiebreak. If neither flag has fallen by the turn cap (**20 turns** on the shipped scenario — the cap is per-scenario data, set in the scenario file's turnCap field, §2.13.2 and §4.7 Stub 7), the match resolves as a battle of attrition. The full procedure is one guard, one three-key comparison, and one grade. Every input is a value the game already tracks for the economy (§2.7) and already shows on the standings scoreboard (§2.11): the tiebreak adds no new state, only an ordering over existing state.

Tally definition. A side’s **combat Fame** is the Fame it earned from unit kills and undamaged-strike bonuses (§2.7). It **excludes** passive factory and town income — counting income would let a staller re-earn the turtle exploit (§1.5 #1) by sitting on objectives. The flag bonus (+500) can never appear in a capped tally: destroying the flag ends the match immediately, so no match that reaches the cap contains one.

Resolution procedure, in order:

1. **Mutual-passivity guard.** If *both* sides’ combat Fame is zero — nobody engaged — the match is an immediate **draw**. It does not fall through to the keys below, because “objectives held” would otherwise re-crown a turtle who simply sat on more factories.
2. **Lexicographic comparison** — higher wins at the first key that differs:
 1. **Combat Fame earned.** The anti-turtle lever (PX-4): a player who sealed a corner and refused to fight scores zero here and loses the cap to anyone who dealt damage. (The guard already covers the both-zero case, so no separate “both sides fought” precondition is needed on the later keys — if this key ties at a nonzero value, both sides fought by construction.)
 2. **Objectives held** — the factories and captured towns a side owns at the cap, as *X of N*. Ownership only: a capture in progress (§2.7) counts for nobody until the objective flips. Towns count here even though the domination win is factories-only — domination must stay hard-won; this key merely breaks an exact Fame tie. With ~4 factories plus towns on the map (§2.3, §2.7) this is a genuine spread.
 3. **Surviving strength** — total remaining HP of a side’s units. Deliberately last: alone it would re-reward turtling, but by the time it applies both sides have earned identical nonzero combat Fame, so an HP edge measures trading efficiency between two sides that both fought. It is free to compute (the scoreboard already sums it) and decides matches that would otherwise be draws.
3. All three keys equal → **draw**. Acceptable for the prototype; a competitive build would add a sudden-death overtime.

Victory quality. A result is graded as a **tier**, not a number:

Tier	Trigger
Decisive win	Enemy flag destroyed, or territorial domination
Marginal win	Led the attrition comparison at the cap
Draw	The passivity guard fired, or all three keys tied

Tiers rank categorically: a **decisive win always outranks a marginal win**, regardless of how much Fame either side piled up. Combat Fame is only the sort key *inside* the tiebreak, never the grade — keeping the two separate prevents a long capped match’s accumulated kill-Fame from “outscore” an actual flag kill (§1.5 #5). For self-play tuning, log the tier plus the Fame breakdown per match. The result: a flag kill is always the best outcome, so Pillar 2 (“short and decisive”) is enforced by the grading itself, not just by the turn cap.

Invariants (each phrases directly as a T-CAP- test):

1. **T-CAP-01** — Flag destruction on any turn at or before the cap yields a Decisive result; the tiebreak procedure is never evaluated.
2. **T-CAP-02** — Both sides at zero combat Fame at the cap → Draw, regardless of objectives held or surviving HP.
3. **T-CAP-03** — Passive income never decides the cap: a side holding 4 factories with zero kills loses the cap to a side holding 1 factory with one Infantry kill.
4. **T-CAP-04** — No capped tally contains the +500 flag bonus.
5. **T-CAP-05** — A capture in progress at the cap contributes zero to “objectives held” for either side.
6. **T-CAP-06** — Tier order is Decisive > Marginal > Draw for *any* pair of Fame totals.
7. **T-CAP-07** — Determinism: identical end state at the cap → identical result and tier.
8. **T-CAP-08** — Controlling every factory at the start of your turn ends the match immediately as a Decisive win; towns do not count toward domination.

Why this shape (the delete-test). Every piece of the apparatus was tested by deletion. Delete key 1 → the documented turtle exploit (§1.5 #1) returns whole. Delete the guard → a four-factory turtle beats a one-factory turtle without a shot fired. Delete key 2 → contesting the neutral factories (§2.7’s mid-game prize) stops mattering at the cap. Delete key 3 → more draws for zero savings, since the HP sum already exists for the scoreboard. Delete the tiers → a capped grind’s tally can read as “beating” a flag kill in tuning logs — the inversion §1.5 #5 closed. Delete the domination backstop → a walled-in flag forces every such match to run the full cap for a Marginal result instead of ending early and Decisively (Pillar 2). The one piece that failed the test — the floated per-turn Fame decay — is cut in this revision (§1.6 row 5). What remains is a guard, a sort, and an enum, all over state the game tracks anyway.

On the turn cap vs. real time. The cap bounds *turns*, which guarantees the match terminates; it does not bound wall-clock minutes. Against the shipping single-player AI (which moves instantly) that is a non-issue, so “~20 minutes” (§1) is an expected duration at a normal move pace, not a hard real-time ceiling. The only mode that would need a per-turn timer is **PvP hotseat**, a stretch feature (§2.10) — if it ships, add a move clock there.

2.9 Opponent AI (runtime gameplay system)

- **Baseline (ships): simple objective-seeker.** Two phases each turn, so the AI actually *uses* the economy (§2.7), not just the map:
 - **Economy phase.** At each factory it holds: if it can afford a unit, build one from a default buildlist (mostly Infantry, an occasional Tank), spawning per §2.7. It spends Fame and replaces losses instead of hoarding.
 - **Unit phase, per unit:** (1) an idle **Infantry** adjacent to or near an uncaptured, **undefended** factory/town moves onto it to capture — the AI contests objectives, keeping capture, production, and the “objectives held” tiebreak live on *both* sides; (2) if an enemy is within reach after moving, attack — prefer the enemy flag, else the best expected-damage target; (3) ranged units (Artillery) fire from maximum standoff so they don’t eat a counter; (4) otherwise advance along the cheapest path toward the enemy flag.
 - One cheap guard keeps it from looking broken: **skip a strictly-losing attack** (the unit would die and trade down).
 Deliberately un-clever — decisive, readable, and fully testable. This is the shipping opponent.
- **Difficulty = a starting-Fame handicap, not a smarter AI** (mirrors *Conflict*). The baseline routine is identical at every tier; only the economy shifts: **Easy** = player +150 opening Fame; **Normal** = even (200/200, §2.7); **Hard** = player −100. Deterministic and trivially tunable, with no AI-quality risk. (*Stretch dial, documented only: also scale AI income ±25%/tier*)
- **Stretch: AI second pass (utility + threat map).** Upgrade the baseline to score candidate actions on attack value (expected damage vs. counter), objective proximity, and safety (a threat map of enemy reach). Week-3 work, only if the baseline proves too exploitable in self-play.
- **Stretch: LLM commander.** Serialize the board to text, enumerate legal moves, ask the model to rank one, then **validate and apply** (never trust an unchecked move). Behind a toggle, heuristic as fallback. This is the one place an AI *agent* appears in the shipped product — see §3.

2.10 Scope table

	Contents
IN (core; phased wk 1–3 per §4.4)	Grid; 4 units; 6 terrains + the Factory tile ; move + attack; multiple factories (home-per-side + contested neutrals, ~4) + capture + Fame production (<i>these land wk 3, not wk 1–2</i>); heuristic AI; one hand-built scenario ; win-by-flag; functional UI
STRETCH (if ahead)	2nd–3rd scenario; LLM commander; fog/recon; sea/air units; map-gen MCP toolset; 2-player hotseat
CUT	All 16 original scenarios; campaign/meta; zones of control; elaborate art; anything real-time

2.11 UI/UX

Stratocracy’s interface has one job: make a deterministic game *look* deterministic. Every rule in §2 is knowable before the player commits, so the UI’s standard is not “pretty” but **no surprises** — if the player is ever surprised by an outcome, the UI failed, not the player. Three principles govern everything below:

1. **Hover is information, click is commitment.** Hovering never changes game state; it only reveals (terrain lines, path previews, the forecast). A single left-click commits. Cancel is always right-click or Esc.
2. **Every element earns its pixels.** Each HUD element is audited against the decision it supports (§2.11.2); anything supporting no decision is cut. Two deliberate cuts up front: **no minimap** (the one shipped scenario is a small skirmish map, §2.7 — it fits on screen at default zoom; pan/zoom covers the rest) and **no combat log** (determinism plus the forecast make a history redundant for the prototype; a log rides the replay format if that ships, §4.10).

3. **Rules-critical information is never transient-only.** Toasts and banners are receipts; the durable version of every fact lives on a persistent or hover-recallable surface.

UI is budgeted as core work from week 2 (§4.4, §4.5 — “UI underestimated” is a named risk) and everything below is scoped to what one person can build in UMG on top of agent-scaffolded widget skeletons (§3, UI Scaffold role).

2.11.1 Control scheme

Selection state machine. The core loop (§2.1: select → move → act → done, any unit order) maps to four UI states:

```

IDLE —LMB on own unacted unit→ SELECTED (reachable hexes lit, §2.5)
IDLE —LMB on own factory→ PRODUCTION MENU (§2.11.5)
SELECTED —LMB on lit hex→ MOVED (attack targets lit; Wait
available)
SELECTED —RMB / Esc→ IDLE (nothing committed)
MOVED —hover enemy target→ forecast card shown (§2.11.3)
MOVED —LMB on lit target→ attack resolves exactly as forecast →
unit DONE
MOVED —Space (Wait)→ unit DONE without acting
MOVED —RMB / Esc→ unit DONE where it stands (move already
spent)*

```

* Under a ruling that grants move-undo (**Q11**, §4.7 — currently unruled, and no gate assumes it), RMB/Esc in MOVED instead reverts the unit to its pre-move hex and returns to IDLE. Until that rule is adjudicated, the shipping semantics are the conservative ones shown: a completed move stands. The two behaviors share a UI; only the rules module differs.

Capture and build need no extra verbs. Capture is by presence (§2.7: an Infantry that ends its move on a capturable tile begins capturing — a progress pip appears, no button). Building is the factory’s own interaction, not a unit’s. This keeps the per-unit action vocabulary to exactly two: *attack* or *wait*.

Input reference.

Input	Effect
Hover hex	Terrain line in the info panel (§2.11.2); if a unit is SELECTED, dotted path preview along the cheapest route (§2.5) with the terrain-cost tick per hex
Hover unit	Unit stats in the info panel; if own MOVED unit has it in reach, the forecast card (§2.11.3)
LMB	Select own unit / commit previewed move / commit forecast attack / open production menu on own factory / activate buttons
RMB / Esc	Cancel: close menu or forecast, deselect, back out one state (see machine above)
MMB drag or WASD / arrows	Pan camera
Mouse wheel	Zoom (two or three fixed steps, not continuous — readability over cinematography)
Tab	Cycle to the next unit that has not acted
Space	Wait — mark the selected/moved unit done without acting
F	Snap camera to your flag unit
B	Open the production menu at the selected owned factory (or your home factory if none selected)
Enter	End turn — with a confirm dialog if any unit has not acted: 3 units have not acted. End turn?
Z	Undo move (<i>only if Q11 (§4.7) is ruled to grant undo; otherwise unbound</i>)
Any click / Esc during enemy turn	Skip AI playback to its end state (§2.11.2, turn banner)

No hidden double-functions, no drag-to-move, no context menus. A first-session player can complete a match knowing only: hover, LMB, RMB, Enter.

2.11.2 HUD layout & information architecture

Screen layout.

+-----+ [DIRECTIVE STRIP / TURN BANNER] +-----+			
TURN 12 / 20	(top center, transient)		FAME 350
-----			+175/turn
YOU ENEMY			+-----+
Destr. 450 600			
Obj. 4/8 3/8			
HP 47 55	(hex battlefield)		
+-----+			
(live scoreboard,		[H] flag markers on-map	
§2.11.4)		[forecast card floats near target]	
+-----+			
INFO PANEL	[toast queue,	3 units idle	
Woods	bottom center]		[END TURN ↶]
move 2 · def +20%			+-----+
+-----+			
+-----+			
-+			

Three information layers, strictly tiered:

1. **Persistent** (always on screen): scoreboard + turn counter, Fame pool + income rate, End Turn + idle-unit count, on-map flag H markers and unacted-unit pips. These are the four standing decisions of every turn — *am I winning the cap, can I afford to build, is my turn actually finished, what must I protect*.
2. **Contextual** (selection-driven): reachable-hex highlight, path preview, attack-target highlight, info panel content, forecast card, production menu, capture pips. Appears with a selection or hover, gone when it ends.
3. **Transient** (event receipts): toasts (+100 Fame – Factory, +[N] HP – repaired, +150 Fame on a kill), turn banner, one-shot onboarding lines (§2.11.6), cap-approach banners (§2.11.4). Each transient fact has a durable home: income toasts restate what the Fame widget's +X/turn already shows; kill toasts restate the scoreboard's Destroyed row.

Earn-your-pixels audit — every persistent/contextual element, the decision it supports, and the rule it surfaces:

Element	Decision it supports	Rule surfaced
Scoreboard (turn + 3 rows)	Force combat or hold; how close is the cap	§2.8 tiebreak order, turn cap
Fame pool + +X/turn	Build now vs. save; which neutral factory is worth a fight	§2.7 income (+100 factory / +25 town), costs
End Turn + idle count	Is my turn genuinely spent	§2.1 per-unit loop
Flag H marker (both sides, always visible)	What to protect, what to hunt	§2.4 flag death ends the match; <i>Conflict's</i> H convention
Unacted pip on own units	Which units still have a move	§2.1
Reachable-hex highlight	Where can this unit truly go	§2.5 — “the real move set, not an estimate”
Path preview with cost ticks	Which route; exposure en route (e.g. a turn spent on the Bridge at −10%)	§2.3, §2.5 cheapest-route
Attack-target highlight (incl. Artillery’s dark range-1 hole)	Who is actually hittable from here	§2.4 ranges; the Artillery dead zone
Forecast card	Commit this attack or not	§2.6 — the whole section (§2.11.3)
Info panel	Speed-vs-cover terrain trade; matchup reading	§2.3 table, §2.4 stats
Production menu	What to buy, where it will spawn	§2.7 build & spawn rules (§2.11.5)
Capture progress pip	Hold or abandon the capture	§2.7 capture over N turns
Repair eligibility pip (owned tile, damaged unit selected, no adjacent enemy)	Retreat to heal or keep fighting	§2.7 repair + anti-fortress clause

Info panel (bottom-left, hover-driven, ~3 lines, never modal): - Hovered hex: terrain name, move cost, defense bonus, and status if capturable — `Factory · move 1 · def +15% · yours (+100/turn) or · neutral or · enemy`. - Hovered unit: name, HP as 12/20, Atk/Def/Move/Range, has acted flag. The flag unit’s panel is red-edged and appends `FLAG – its loss ends the match`. - Empty when nothing is hovered. It never covers the board’s lower-center where fighting happens.

Turn banner & AI playback. A brief `YOUR TURN / ENEMY TURN` banner marks the IGOUGO handoff (§2.1). The headless AI resolves instantly (§2.8); the presentation layer **replays its action list at a watchable fixed pace** (~0.5 s per action, camera stepping to each) so the player can read what the AI built, captured, and attacked — this is presentation pacing only, no rules change. Any click or Esc skips to the end state. First-session value: watching the AI’s economy phase is how the player learns the enemy shares the same Fame economy (§2.9).

2.11.3 The attack forecast — the game’s centerpiece display

Combat is a pure function (§2.6, §3 spec), so the forecast is not an estimate — it is the resolution, shown early. It appears on **hover** over any lit target from the `MOVED` state; **LMB commits**, **RMB/Esc cancels**. The card:

```
+-----+
| ATTACK FORECAST                               |
| Artillery → Tank      (Woods +20%)            |
|                                     |
| You deal    3 dmg    Tank 20 → 17            |
| Counter     0              out of range      |
|                                     |
| [LMB] Commit           [RMB / Esc] Cancel    |
+-----+
```

(Values follow the §3 formula: *Artillery atk 10 at full HP vs Tank def 5 in Woods* → $\text{round}(10 \times 1.0 \times 0.8) - 5 = 3$; counter 0 because the attacker sits outside the Tank’s range 1.)

Hard rules for the card:

- **The defender’s terrain bonus is named inline** (`Woods +20%`), every time — terrain defense must never read as hidden dice. (Per the §3 invariant, it is always the

defender's hex; the card's placement of the modifier next to the defender teaches that for free.)

- **The counter line is never omitted, and always states its reason:** a number, out of range, or defender destroyed. This line is where the positional RPS triangle (§2.4) becomes visible — there is no type chart to consult, deliberately, so Counter 0 – out of range is the Artillery-beats-Tank lesson and Counter 5 on an adjacent brawl is the Tank-beats-Recon lesson.
- **HP is shown before → after** for the defender (and for the attacker whenever the counter is nonzero), so the HP-scaling term of the formula (§2.6 — a damaged attacker hits softer) is observable across fights rather than asserted.
- **Lethal forecasts state their reward:** if the defender dies, the card appends Destroys Tank +150 Fame (kill ≈ half cost, §2.7) — the tiebreak's combat-Fame criterion (§2.8) is priced on the commit card, not discovered at the cap.
- **Flag warning band:** if the forecast is lethal to either flag, the card gains a band — FLAG AT RISK – this attack ends the match (own flag: red; enemy flag: gold, with +500 · Decisive victory). No player can end a match, theirs or the enemy's, without having been told on the card they clicked.
- **Determinism is restated once,** via the first-attack one-shot (§2.11.6): Check the forecast. It is exact – what you see is what resolves. After that the card carries no reassurance text; the outcomes matching the card, every time, are the proof.

Selecting Artillery renders its attack ring as range 2–3 **with the range-1 hole drawn visibly dark** — the dead zone on the map is the “Recon runs it down” lesson in pixels, using the attack-target highlight already budgeted.

2.11.4 The live Fame scoreboard

The scoreboard exists because of revision §1.5-#1: the tiebreak must never be a hidden win condition. It is persistent (top-left), compact, and its **rows are ordered top-to-bottom in exact tiebreak order (§2.8)** — the layout is the rule, read passively all match:

TURN 12 / 20			
	YOU	ENEMY	
Destroyed	450	600	◀
Objectives	4/8	3/8	
Unit HP	47	55	

- **Turn counter** against the cap, always. (/ 20 is *Ferrum Crossing's* cap, §2.13.2; the cap is per-scenario data, so the widget reads turnCap from the scenario rather than hardcoding a number.)
- **Destroyed** = combat Fame earned (kills + undamaged-strike and flag bonuses, §2.7) — **passive income is excluded**, exactly as the tiebreak excludes it. Hover tooltip: Fame from kills and strike bonuses. Factory income does not count at the cap. This row deliberately does *not* equal the spendable Fame pool (top-right widget), and the tooltip on each names the difference — the one place the single-currency design (§2.7) needs a disambiguating sentence.
- **Objectives** as *X of N* over all factories + capturable towns (§2.8 criterion 2), *N* supplied by the scenario (§2.13) — **N = 8** on *Ferrum Crossing* (4 factories + 4 towns), as the mock shows.
- **Unit HP** = surviving strength (criterion 3), listed last because it is last.
- A **chevron (◀)** marks the current attrition-tiebreak leader, evaluated in criteria order, and flips visibly when the lead changes. It is drawn beside the leading side's value — in the mock above the enemy leads at criterion 1, 600 combat Fame to 450 (§2.8: higher wins), so the chevron sits on the enemy column. If both Destroyed values are zero, the chevron is replaced by – no engagements – spanning the row: the mutual-passivity draw (§2.8) made visible before it bites.
- **Cap-approach banners** (transient, once each): at cap–5, 5 turns to the cap. The scoreboard decides a capped match.; additionally, if both sides are still at zero combat Fame, No engagements. A capped match with no combat is a draw.

End-of-match screen. The result is the tier first (§2.8 — Decisive / Marginal / Draw), then the same three rows in the same order, so the verdict is always a restatement of what was on screen all match. Beneath the tier, one **faction-voiced result line** (per the setting guide: faction voice appears only on result screens; ≤ 30 words; field-manual register). Samples, one per case, generated content to follow these:

- Directorate, decisive: Command directive fulfilled. The enemy flag is struck from the record. Order is restored.
- Directorate, marginal: The cap is reached. The ledger favors the Directorate. The record stands.

- Vanguard, decisive: Their flag is down. We hold the ground. That's the whole report.
- Vanguard, marginal: Cap hit. We did the damage; they held the rear. The ground says we win.
- Draw, neutral system voice: Turn cap reached. Attrition equal. Recorded as a draw. / mutual passivity: Turn cap reached. Neither side engaged. Recorded as a draw.

2.11.5 Production menu & match-flow surfaces

Production menu — opens on LMB on an own factory (or B), anchored beside it:

FACTORY — BUILD		Fame: 250	
Infantry	100	[Build]	
Recon	150	[Build]	
Artillery	200	[Build]	
Tank	300	----	(need 50)
Spawns here, or adjacent if occupied.			

- Unaffordable rows are greyed with the shortfall named (need 50 at the mocked 250-Fame pool against the Tank’s 300 cost, §2.4), never hidden — the price list is also the strategy lesson (§2.4 costs).
- The spawn rule (§2.7: factory hex if free, else an adjacent free hex) is one static line in the footer. If the factory is fully boxed in, the footer swaps to Boxed in – build waits for a free hex. and Build buttons disable: the space-throttle (§2.7) explains itself at the moment it applies.
- When any unit is affordable and the factory has not built this turn, the factory tile shows a small BUILD pulse — the nudge that connects hoarded Fame to an army (a first-session failure mode, §2.11.6 ledger).
- The row hover shows the unit’s stat line in the info panel, so buying is done with the §2.4 table in view.

Pre-match: the static briefing overlay (three callouts: flag, Bridge, factories — §2.11.6-A).
Post-match: the end screen above. That is the complete screen list for the prototype: title/menu, briefing, match, result. No settings screen beyond volume + resolution is budgeted (Enhanced Input remap is a polish item).

2.11.6 Onboarding — the first session

Philosophy. No manual, no tutorial mode, no modal text walls. Three teachers:

1. **Constraint** — the first turn removes every option except the one being taught.
2. **The forecast** (§2.11.3) — deterministic combat means every attack is a free, truthful lesson in terrain defense, range, and HP scaling. The plan makes the player *read* the forecast once, early; after that the game teaches itself.
3. **One-shot event tips** — a single system-voice line (≤ 30 words, tone bible), fired the first time a concept becomes relevant, never repeated (boolean flags in the save slot, §4.1).

The first match runs on the one shipped scenario at **Easy** by default (player +150 opening Fame, §2.9) with a **guided opening**: four scripted directives inside a fixed four-turn window — the first appears on turn 1, the strip and every beat behind it are gone for good at the end of turn 4, then hands-off. Any completed match on the save skips all guidance automatically; a skip guidance control kills it instantly for anyone, and kills the guided opening’s board state with it — the objective ring and the turn-1a unit marker clear in the same frame as the strip.

A. Pre-match briefing — a dimmed board, three anchored callouts, click-through (~5 s): **your flag** (If it falls, you lose. It cannot be rebuilt.), **the Bridge** (The only land crossing.), **factories** (Hold them for Fame. Fame builds units.). Why only three: the flag is the win condition, factories are the economy, and the Bridge is the map-reading trap — a player who misses the single Water crossing misplans the whole match.

B. Guided opening (turns 1–4), via a one-line directive strip at top center, one instruction at a time, each retiring on completion:

Turn	Constraint	Directive	Teaches	Retires when
------	------------	-----------	---------	--------------

Turn	Constraint	Directive	Teaches	Retires when
1a	Only one marked Infantry selectable, others dimmed (hover: Locked this turn.). End Turn is inert until that Infantry has moved (hover: Move the marked Infantry first.) — this is what makes 1a retire inside turn 1 in every branch, and it is the only guided-opening constraint that gates a player <i>input</i> rather than a selection, adopted under Q27 (§4.7), ruled — it was registered rather than assumed because it gates an input	Select the marked Infantry. Lit hexes are its true reach. Click one to move.	Selection; the highlight is the real move set (§2.5)	Move completes
1b	End Turn pulses	End turn. The enemy moves; then you.	IGOUGO (§2.1) — the player watches a full AI turn	Enemy turn ends A capture pip appears — on whatever turn that happens. <i>Standing</i> means it stays outstanding , not that it holds the strip. How many turns it actually holds the line is decided by rules 1–2 and by when the pip lands, and the schedule table’s three branches are exhaustive: twice — turn 2, then a turn-4 rule-2 last call, if the pip has still not landed (wandered); once — turn 2 only, retiring on a turn-2 or turn-3 pip; or never — the pip lands on turn 1 and beat 2 retires before rule 1 can select it (fast lane). On every turn it does not hold the line it runs on the ring and the unit marker. Hard-expires at end of turn 4
2 (standing)	None on selection. The scenario’s designated neutral factory (guidedOpening.objective, §2.13.1) is ringed from turn 1; its info-panel line appends Only Infantry captures.	Move the Infantry onto the ringed Factory. Only Infantry captures.	Capture; the Infantry-only rule (§2.7)	once — turn 2 only, retiring on a turn-2 or turn-3 pip; or never — the pip lands on turn 1 and beat 2 retires before rule 1 can select it (fast lane). On every turn it does not hold the line it runs on the ring and the unit marker. Hard-expires at end of turn 4
	None. Fame ≥ 100	Spend Fame at your	Fame →	A unit spawns — on whatever turn

3	guaranteed by 200 start + home income	Factory. costs 100.	factory → unit	that happens, including
Turn	Constraint	Directive	Teaches	Retires

The Turn column is the beat's **order index** — the turn it takes the line in the common case — not a floor. Rules 1–2 below assign the line, so a beat moves up whenever a lower-numbered beat has already retired: beat 3 takes turn 2 in the fast-lane branch of the schedule table. The strip shows **one directive at a time**, and the line is assigned at the start of each turn by two rules in order:

1. the lowest-numbered **outstanding** beat that has **not yet held the line on an earlier turn**;
2. if every outstanding beat has already had its turn on the line, the lowest-numbered outstanding beat — a **last call**.

A beat gives up the line either the instant it retires (which is how 1a hands off to 1b inside turn 1) or at the end of the turn it first appeared, whichever comes first. **Giving up the line is not retiring.** An outstanding beat that has yielded keeps running on its board state: for beat 2 that is the ringed objective and the marked Infantry, which are the same instruction in spatial form and have been on screen since turn 1. The text line introduces the objective; the ring is what reminds.

The consequence that matters: **no beat expires unheard**. Each beat either retires on its own event — its lesson landed in play, which is exactly what beat 2 does on turn 1 in the fast-lane branch, before it has ever held the line — or it holds the line at least once before the window closes, which is what rule 2 exists to force. A beat 2 that hangs — the *Ferrum Crossing* 1-MP-slack case below — cannot starve beat 3, whose Fame → factory → unit lesson is the one the \$2.11.6-D ledger confirms with a bought unit on the board. Rule 2 then makes turn 4 a last call for whatever is still outstanding, so the strip's final turn is spent on the player's actual gap rather than on order-of-arrival:

Turn	Common case — pip lands turn 2	Wandered case — pip lands turn 3 or 4	Fast lane — pip lands turn 1 (<i>The Causeway</i> , 3 MP lanes)
1	1a, then 1b when 1a retires	1a, then 1b	1a, then 1b; the pip lands inside this turn, so beat 2 retires without ever holding the line
2	beat 2 (rule 1) — retires on the pip	beat 2 (rule 1) — holds, then yields	beat 3 (rule 1)
3	beat 3 (rule 1)	beat 3 (rule 1)	beat 3 — rule 2 last call, untagged
4	beat 3 — rule 2 last call, tagged	beat 2 — rule 2 last call, tagged	beat 3 — rule 2 last call again , tagged
end of 4	strip gone; anything still outstanding expires	strip gone; anything still outstanding expires	strip gone; anything still outstanding expires

How to read a cell. Each cell names the beat that rules 1–2 select for that turn, given the column's pip timing *and* beat 3 still outstanding when its cell is reached. Beat 3 retires on a spawn **event**, not on turn 3, and Fame ≥ 100 from turn 1, so a player who builds early retires it early. Every cell is therefore read as: *this beat — or, if it has already retired, the next beat rules 1–2 select — or nothing, if none is outstanding*. Turn 1 is identical in all three columns because 1a and 1b cannot outlive it: End Turn stays inert until the marked Infantry moves (beat table, row 1a), and 1b's retire condition **is** the turn boundary.

Is the strip ever quiet before the end of turn 4? No — there is no live-but-blank strip in this system. Rule 2 has no exit: while any beat is outstanding, some beat holds the line. The only empty state reachable before end of turn 4 is the strip being **gone**, and its condition is exact — *all four beats retired*: 1a and 1b inside turn 1, beat 2 on its pip, beat 3 on a spawn. The earliest that can occur is the end of turn 1, and only in the fast-lane branch with a turn-1 build; it is permanent, per the disappearance rule below, not a pause. In particular, fast-lane turn 4 is **not** unconditionally quiet: a player who has not bought a unit still has beat 3 outstanding, and rule 2 returns it to the line, exactly as the common-case column does at its own turn 4. That is the guarantee working, not an exception to it.

A **turn-4** last-call line is the beat's own text with a dim right-hand tag, so the player is never dropped mid-instruction without warning. The tag states a fact about the *window*,

not about rule 2, so it renders on turn 4 only: an earlier rule-2 last call (fast lane, turn 3) shows the same line untagged, because guidance does not end that turn and the strip must not say it does:

```
+-----+
-+
| Move the Infantry onto the ringed Factory. Only Infantry captures.
|
|                                     guidance ends this
turn|
+-----+
-+
```

The strip disappears for good once all four beats have retired, and unconditionally at the end of turn 4; every beat, the objective ring, and the turn-1a marker expire with it.

Why beat 2 is a standing directive, not a turn-2 deadline. Its target is guaranteed by §2.13.1's **opening-capture reachability** invariant: for each seat, at least one Infantry deployment hex has a Bridge-free land path to a **neutral factory** costing ≤ 6 movement points — two turns at Infantry Move 3 (§2.4) — and the scenario file names that unit and that factory in `guidedOpening.infantry` and `guidedOpening.objective`. The directive strip reads exactly those two fields: `guidedOpening.infantry` is the Infantry marked in beat 1a, `guidedOpening.objective` is the factory ringed from turn 1. Nothing is measured at runtime and no “nearest objective” heuristic is used — the lane is authored, machine-validated, and recorded as a number by `validate_scenario` (§4.2), so the onboarding and the map can never disagree about which factory the player was told to take.

What the invariant does *not* buy is a safe turn-2 deadline, and that is why beat 2 retires on an event rather than a turn number. Beat 1a hands the player a free move in any direction — that is its whole lesson, and the onboarding must not punish the player for using it. On *Ferrum Crossing* both lanes cost **5 movement points**, not 5 and 4 hexes: West (1,5) → South (5,7) is 5 hexes of Plains at cost 1, and East (9,3) → North (6,2) is 4 hexes but one is a mandatory Woods ring hex at cost 2 (§2.3). Against the 6 MP budget that is **1 MP of slack**, the tightest of the three maps and the only one that is tight at all: both stretch maps were redrawn on even row counts (*Longwater March* 13×8 , *The Causeway* 9×8 , both $\text{rot}180$), and their lanes price at **4 MP / 4 MP** and **3 MP / 3 MP**, so they carry **2** and **3 MP** of slack respectively (§2.13.1's table). No lane on either 8-row map carries 4. A single turn-1 step spent walking off the lane therefore pushes the pip to turn 3 — and a hard turn-2 retire condition would strand the strip on a directive the player had already been made unable to satisfy that turn. The standing directive absorbs precisely that: on *Ferrum Crossing*, where a turn-1 pip is impossible at 5 MP per lane against Move 3, it takes the line on turn 2, stays outstanding until the pip appears, and hard-expires at the end of turn 4 — the last turn of the guided window, not one turn past it. It persists as an *objective*, not as a line of text: it holds the strip for turn 2, yields the line to beat 3 for turn 3, and returns for a turn-4 last call only if the pip still has not landed. That is **two turns on the line at the outside** — one in the common case, where the turn-2 pip retires it, and none at all on a map whose lanes let the pip land on turn 1 (the fast-lane column). That is the whole price of “standing”, and it is paid in ring and marker rather than in strip time, so the Fame → factory → unit lesson is never the thing that gets crowded out by a slow walk. Ringing the objective from turn 1 biases beat 1a's free move onto the lane without constraining it, so in the common case the slack is never spent and the pip lands on turn 2 as designed.

“Capturing” by turn 2, never “captured.” The invariant promises the Infantry is *standing* on the factory at the end of turn 2, not that the tile is yours: under Q4's $N = 1$ reading the tile flips at the start of turn 3 (§2.7). So the directive reads **Move the Infantry onto the ringed Factory** — never “capture it” — and it retires on the **capture pip**, the arrival receipt, not on the ownership flip. The flip gets its own confirmation one turn later via the **+100 Fame – Factory** toast, which is where the concept ledger's Capture row already ends. This wording is also correct if N is ever ruled 2: the pip is the arrival event either way.

C. Event-driven one-shots (once each, at first relevance): first attack hover → Check the forecast. It is exact – what you see is what resolves.; first Artillery strike at range 2–3 → No counter at this range. Artillery strikes beyond reply.; first Recon-vs-Artillery forecast → Recon closes fast. Artillery cannot answer at range 1.; first Tank-vs-Recon forecast at range 1 → Armor wins the adjacent fight.; first Water hover with a unit selected → Impassable to land. Cross at the Bridge.; first Bridge hover → The only crossing. –10% defense – do not stall on it.; first combat Fame → +150 Fame. Kills count at the cap. Income does not. (*amount = actual award*); first undamaged strike → +[N] Fame – undamaged strike.; first repair blocked by adjacency → No repair – enemy adjacent. Break contact to repair.; first repair tick → +[N] HP – repaired at Factory.; plus the two cap-approach banners (§2.11.4).

D. Concept ledger — every concept, its failure point, its teach moment, its confirmation:

Concept	Where a first-timer gets lost	Taught at	Confirmed landed when
Hexes & movement	Clicks the map with nothing selected	Turn 1a: single selectable unit + reach highlight	Move completes; directive retires
IGOUGO turns	Expects simultaneous movement	Turn 1b: pulsing End Turn, then watching the AI's turn	Player ends turn 2 unprompted
Terrain move cost	Lopsided highlight (shallow into Woods/Mountains) reads as a bug	The asymmetric highlight + info-panel hover line	Player routes around a Mountain, or hovers 2+ terrains in one selection
Terrain defense	Same attack, different damage → suspects hidden dice	Forecast names the defender's bonus inline, every time (§2.11.3)	Resolution matches forecast, every time
The Bridge	Reads Water as decoration; “half the map is unreachable”	Briefing callout + Water never lights in any reach set + hover one-shots	First previewed or executed path crosses the Bridge
Forecast / determinism	Veterans hedge for RNG; novices fear committing	First-attack one-shot: <i>It is exact.</i>	Player commits; outcome equals card; usage becomes habitual
Positional RPS triangle	No counter chart exists — the triangle is emergent from range and move, invisible until experienced	The counter-reason line (§2.11.3), the three edge one-shots, the Artillery dead-zone ring	One uncountered Artillery strike lands — the undamaged-strike toast is the receipt
Capture	Parks a Tank on the factory and waits	Turn 2 directive; non-Infantry on a capturable tile shows disabled Capture – Infantry only. Beat 3, which holds the strip on turn 2 or turn 3 in every branch where the player has not already built — and where they have, the lesson landed without it (B); income toasts; BUILD pulse when affordable; greyed rows with shortfall (§2.11.5)	Pip → tile recolors → next turn's +100 Fame – Factory toast
Fame income & build	Fame is abstract; player hoards, never connects factories → army	Turn 2 directive; non-Infantry on a capturable tile shows disabled Capture – Infantry only. Beat 3, which holds the strip on turn 2 or turn 3 in every branch where the player has not already built — and where they have, the lesson landed without it (B); income toasts; BUILD pulse when affordable; greyed rows with shortfall (§2.11.5)	A bought unit spawns and stands on the board carrying its unacted pip — the event , not the directive, so the row also confirms for a player who skipped guidance
Flag = the game	Loses one specific Tank in a “fair trade”; match ends out of nowhere	Briefing callout; persistent H; red-edged info panel; flag warning band (§2.11.3)	Any lethal-to-flag forecast has been shown before any match can end
Turn cap & tiebreak	Assumes “most stuff wins”; income exclusion and passivity-draw are the least guessable rules	Scoreboard rows in tiebreak order (§2.11.4); first-combat-Fame one-shot; cap–5 banners	Chevron flips noticed in play; end screen matches the board watched all match
Repair	Heal silently fails next to an enemy; reads as a bug	Eligibility pip; the enemy adjacent one-shot fires at the <i>blocked</i> case first if it occurs	+ [N] HP – repaired toast at turn start

2.11.7 Art (unchanged)

Flat/low-poly color-coded hexes, simple unit meshes or billboarded icons, generative/agent-assisted. Feel comes from subtle tweens and clear feedback, not VFX. One readability addition, cheap and worth naming: ownership is **double-coded** (faction color *and* a shape/border difference), so faction reading never depends on hue alone.

2.11.8 Build ranking (solo dev, UMG)

Must-have for a playable first session: selection state machine + input table (§2.11.1); reachable highlight and path preview (the path is already computed by §2.5 pathfinding — rendering it is cheap); forecast card with terrain-inline, counter-reason, HP before→after, kill-Fame line, and flag band; scoreboard in tiebreak order with turn counter and chevron; Fame pool + income widget; production menu with grey/shortfall/boxed states; info panel; End Turn + idle confirm; turn banner + paced AI playback with click-to-skip; toast queue; flag H markers and unacted pips; directive strip with the four beats; the one-shot system (string table + boolean flags); pre-match briefing overlay; end-of-match screen with tier + rows + one faction line.

Polish: chevron flip animation; Artillery dead-zone shading (plain target highlight conveys most of it); cap banners' timing tuning; camera smoothing and zoom-to-action; hotkey remap via Enhanced Input; colorblind palette variants beyond the default double-coding; hover-delay tuning; a dedicated micro-tutorial map (explicitly stretch — it would occupy a stretch scenario slot, §2.10).

2.12 Lineage extraction — what we kept, diverged, and cut from *Conflict*

The prototype was digested against the original *Conflict* (Vic Tokai, NES, 1989) manual so the lineage is a deliberate set of decisions, not an accident. **Kept and mapped:** hex board, terrain move/defense, capturable towns/factories, factory production, a single Fame currency, held-objective income, and destroy-the-commander victory. **Extracted as new pillar-fitting features:** the *Bridge* terrain (§2.3), *Repair* at owned objectives (§2.7) — the one survivable slice of *Conflict*'s logistics — *starting Fame* + *difficulty-as-Fame-handicap* (§2.7, §2.9), the *territorial-domination* secondary win (§2.8), and an all-1.0 *type-effectiveness* lever (§3 spec). **Deliberately diverged/cut** (recorded so the choice is on the record, not implemented): the real-time **battle mini-game** (NORMAL/AUTO + weapon/maneuver menus + evasion %) → replaced by the deterministic forecast (Pillar 1); **fuel/ammo logistics** and their supply units → cut for pacing; the **16-map password campaign** → one polished scenario; and *Conflict*'s **combat Fame penalties** (retreat/base/flag) → omitted so a flag kill can never be out-piled by attrition (Pillar 2, §1.5 #5). **Declined levers:** the per-turn **activation cap** (3-Units mode) and **one-type-per-turn** production — both lean against Pillar 2 or duplicate the existing board-space throttle.

2.13 Scenario & map design

All layout values are starter/tuning targets, per the GDD's standing convention (§2.3, §2.7). Nothing here adds a unit, terrain, or rule: the palette is the six movement terrains + the capturable Factory tile (§2.3) and the four-unit roster (§2.4), exactly.

2.13.1 Layout conventions (shared by every map)

- **Coordinates:** (col, row), col 0 = west, row 0 = north; pointy-top hexes (§2.2), odd rows offset +½ hex east (odd-r). ASCII glyphs: p Plains · w Woods · m Mountains · ~ Water · b Bridge · t Town · f Factory.
- **Factories:** each side owns exactly one **home factory** at start; all others are **neutral** (no income until captured, §2.7). The factory set is also the §2.8 territorial-domination win set, so factory count and placement is the primary match-length dial (see 2.13.5).
- **Starting force (standard): 5 units per side — 1 Flag Tank (§2.4, not producible), 2 Infantry, 1 Artillery, 1 Recon** — plus 200 starting Fame (§2.7) ± the §2.9 difficulty handicap. Rationale: one of each producible system is live from turn 1, so the positional RPS triangle and capture are in play before the first build; two Infantry means the standard opening (one to a town, one to a neutral factory) doesn't consume the only capturer; the producible force is worth 550 Fame — about two turns of mid-game income — so losing the opening force is recoverable, not match-ending. *No section outside §2.13 sizes the starting force; this count is pending Director approval (Q15, §4.7), not silently adopted.*

- **Home factory hex starts empty** in every deployment, so the turn-1 build spawns on the factory itself (§2.7 spawn rule) instead of scattering.
- **Validation invariants** (for the §4.2 `validate_scenario` tool — schema per §4.7 Stub 7): every land-passable hex reaches every factory (Bridges are the only Water crossings, §2.3, and there is no sea unit — connectivity is a build-time check, not a hope); all deployment hexes are free and land-passable; factory count in the map file equals the count the domination check uses; declared symmetry is machine-verified **in axial** (§4.7's T-SCN-05 forbids loaded state from holding `(col, row)` at all, so the check has no other place to run). The declarable values are **rot180 or none** — mirror is not one of them, because no odd-r rectangle has a *vertical* mirror axis at any dimension and no map in the set is drawn to the *horizontal* one, which exists only on an odd row count (see the symmetry note at the end of this section); and **rot180** is well-formed only on an **even row count**. Both constraints are **ruled** (Q24, §4.7) and are the shipped rule, not an assumption awaiting one. Whether the horizontal mirror was worth a third enum value — it being available and merely unused rather than impossible — was **answered no at Q26**: the enum stays at two values and T-SCN-10 stays unwritten.
- **Opening-capture reachability** — the invariant the guided opening (§2.11.6-B, turn 2) rests on, and the reason it can cite a scenario guarantee at all. For **each seat**, at least one Infantry deployment hex must have a land path to a **neutral** factory costing $\leq 2 \times$ **Infantry Move = 6 movement points** (§2.4 Move 3) and crossing **no Bridge**: two turns of movement then put that Infantry on the factory hex, and the capture pip appears without making a contested crossing the first lesson. The scenario file names that unit and that factory (`guidedOpening.infantry`, `guidedOpening.objective`, §4.7 Stub 7) so the turn-1a *marked* Infantry is the one already standing on the lane. Measured on the three maps as drawn (cost in movement points, cheapest legal path, Bridge-free):

Map	West lane	East lane
<i>Ferrum Crossing</i>	(1,5) → South (5,7), 5 MP , all Plains	(9,3) → North (6,2), 5 MP (4 hexes, one mandatory Woods ring hex)
<i>Longwater March</i>	(1,2) → (4,1), 4 MP (all Plains)	(11,5) → (8,6), 4 MP (all Plains)
<i>The Causeway</i>	(1,2) → (3,2), 3 MP (via the town at (1,1))	(7,5) → (5,5), 3 MP (via the town at (7,6))

All three pass. Three things the invariant deliberately does *not* promise:

1. **“Capturing by turn 2”, never “captured by turn 2.”** With capture $N = 1$ (§2.7, Q4) the Infantry stands on the factory at the end of turn 2 and the tile **flips on turn 3**. The turn-2 directive therefore retires on the *pip*, which is exactly what §2.11.6-B already specifies.
2. **Slack is not uniform.** *Ferrum Crossing* carries only 1 MP of slack against the 6, so a turn-1 move spent walking away from the lane pushes the pip to turn 3 — still inside the guided window, which runs **turns 1–4** (§2.11.6-B). *Longwater March* carries 2 MP of slack on both lanes, *The Causeway* 3 on both; being symmetric, each carries the *same* slack in both seats, which is what makes them usable as controlled tests. Both stretch lanes are also clear of that seat's own four other starting units. Since the Q22/Q28 fix moved East's second Infantry to (9,1), which lies on none of the eight routes *Ferrum Crossing* prices, **no map's numbers move under either reading of Q21 (§4.7) as drawn** — Q21 is ruled terrain-only, and the shipped map is simply the one where a *future* deployment edit landing in a priced route would flip a gate.
3. **Uncontested, and exactly what that buys.** The designated lane is the seat's *own* neutral — West → South, East → North on the shipped map — and under Q22/Q28 that is a **measured inequality**, not an authorial claim: the opposing seat's cheapest Infantry route to the same objective, ranging over **every** Infantry that seat deploys, must cost **strictly more MP** than the owning seat's lane (T-SCN-11, §4.7). Measured on the set: *Ferrum Crossing* **5 against 6** in both seats, *The Causeway* **3 against 5**, *Longwater March* **4 against 8**.

An earlier draft of this note said “the first lesson is not a race” and stopped there. That was wrong on both counts. It was unmeasured; and on the shipped map as first drawn, East's *second* Infantry at (9,5) tied West's South lane at 5 MP flat — a race under any reading, and the reason §2.13.2's deployment now reads (9,1). What is true, stated at the resolution the number supports:

- **No opposing Infantry can arrive for fewer MP.** That is the whole of the inequality, it holds on all six lanes in the set, and it is the property the gate checks.
- **On the shipped map it does not buy a turn.** 5 MP and 6 MP are *both* two turns at Infantry Move 3 (§2.4). A 1 MP margin is daylight in movement points, not in turns. What converts it into “you get there first” is **player-first IGOUGO** (§2.1): the player’s seat moves before the AI inside every turn, so on the shared arrival turn the player stands on the factory hex first and an opposing Infantry walking the same lane finds the hex occupied and the capture pip already placed.
- **Deployment carries the rest of the load, and is meant to.** After the correction neither East Infantry has a cheaper errand in the south: both sit 5 MP from East’s *own* North objective and 6–7 MP from West’s South one. The southern contest is not merely lost on distance, it is uneconomic — which is a stronger guarantee than the gate can express and a weaker one than the gate can enforce, so the gate keeps the inequality and this note keeps the reasoning.
- **The stretch maps do buy a turn,** and that is the difference an honest note has to show. *The Causeway*’s 3-against-5 is a one-turn lane against a two-turn one; *Longwater March*’s 4-against-8 is two turns against three. The shipped map is the tightest in the set, on slack (1 MP, note 2) and on contest (1 MP) alike, and both facts have the same cause: it is the only map in the set that is not drawn to a symmetry.

Declared symmetry does imply equal lane cost — and that is what makes it worth declaring. A 180° rotation of a hex board is an isometry: it preserves hex distance, and because it maps every hex to a hex of identical terrain it preserves *path cost* too. On a genuinely symmetric map the two seats’ lanes therefore cost exactly the same, and a 1 MP split is not an offset artefact — it is proof the layout is not symmetric. An earlier draft of this section argued the opposite from the odd-r row offset; that confused a storage convention with a geometry. §4.7’s T-SCN-05 forbids loaded state from holding (col, row) at all, so the symmetry check necessarily runs in axial, where no offset exists and a declared symmetry is an isometry or it is nothing.

Two facts about offset rectangles follow, and both stretch maps are drawn to them:

1. **No odd-r rectangle has a vertical mirror axis, at any dimension — but one with an odd row count has a horizontal one.** Even rows occupy columns 0... W-1 and are centred on (W-1)/2; odd rows sit ½ hex east and are centred on W/2. The centres differ by ½ hex for every W, so a single **vertical** axis cannot serve both parities, at any dimension. The horizontal axis is a different question with a different answer: $c \leftrightarrow \square c, r \leftrightarrow \square H-1-r$ is exact whenever H-1 is even — i.e. whenever **H is odd** — because r and H-1-r then share a parity, so every row keeps its ½-hex offset and no column moves at all. In axial it is one parity-free affine map, $\mu(q, r) = (q + r - (H-1)/2, H-1-r)$, integer-valued exactly when H is odd. That is the precise complement of fact 2’s even-row-count precondition, so an odd-r rectangle admits *at most one* of the two symmetries and its row parity chooses which: even H → 180° rotation, odd H → horizontal mirror, never both, never a vertical mirror. 11 × 9 *Ferrum Crossing* (§2.13.2) sits on a geometry that has such an axis and is deliberately not drawn to it (§2.13.4) — an asymmetry that is a design choice, not a geometric accident. Whether a horizontal mirror therefore becomes a declarable value alongside *rot180*, with an odd-row-count precondition, was **answered no at Q26** (§4.7): the enum stays at two values and T-SCN-10 stays unwritten. No map in the set declares one, and a later ruling could add it purely additively.
2. **An odd-r rectangle admits a 180° rotation only when the row count is even.** The rotation pairs row r with row H-1-r. With H odd those two rows share a parity, so each must land on a row carrying the same ½-hex offset — and a rotation is one rigid motion with one centre, while fact 1 has just shown the two parities’ centres differ by ½ hex. Put that centre on the even rows, at (W-1)/2, and every odd row reflects as $c \leftrightarrow \square W-2-c$, which sends column W-1 to column -1. Put it on the odd rows, at W/2, and every even row reflects as $c \leftrightarrow \square W-c$, which sends column 0 to column W. Either placement throws one column of every second row off the board. In axial the same fact is pure arithmetic: the rotation constant $W - H/2$ is a half-integer on odd H, so *no* hex has a hex image — on 9 × 9 the constant is 4.5 and (1,1) rotates to column 6.5, which is why §4.7’s T-SCN-09 refuses the file with a reason instead of reporting failed comparisons. With H even the parities alternate and the board closes exactly under $\mu(c, r) = (W-1-c, H-1-r)$. This is why *Longwater March* is 13 × 8 and *The Causeway* is 9 × 8 rather than nine rows each.

The validator still **measures and records each seat’s cost as a number** (T-SCN-08) rather than reading it off the symmetry flag. Not because the flag is untrustworthy in principle, but because it is an *authored declaration* and measurement is the only thing

that catches it being wrong. That is exactly what it caught here: the flag said mirrored, the numbers said 3 and 4, and the numbers were right.

2.13.2 The shipped scenario — *Ferrum Crossing* (§2.10 IN)

Spec	Value
Dimensions	11 × 9 = 99 hexes
Starting units per side	5 (standard force, 2.13.1)
Factories	4 — one home per side + 2 neutral (verbatim the §2.7 “~4 total” layout)
Towns	4
Turn cap	20 turns (the §2.8 per-scenario cap, Stub 7 turnCap)
Symmetry	Asymmetric — handicap story below
Estimated match length	12–16 turns (reasoning below)

Layout.

```

r0:  p p p p p ~ p p p p p
r1:  p p p T p B w p T p p
r2:  p p w p p ~ F w p p p
r3:  p p p w p ~ w p p p p
r4:  p F p p p B w w p F p
r5:  p p w p p ~ p p p p p
r6:  p p p p m p m T p p p
r7:  p p T p p F p p p p p
r8:  p p p p p p p p p p p

```

Key coordinates. Water (5,0)(5,2)(5,3)(5,5) · Bridges **(5,1)** north, **(5,4)** center · Home factories: West **(1,4)**, East **(9,4)** · Neutral factories: **North (6,2)** on East’s bank, **South (5,7)** in the mountain pass · Towns (3,1)(8,1)(2,7)(7,6) · Pass mountains (4,6)(6,6) · Woods bands: East bank (6,1)(7,2)(6,3)(6,4)(7,4), West approaches (2,2)(3,3)(2,5).

Terrain distribution (99 hexes): Plains 75 · Woods 8 · Mountains 2 · Water 4 · Bridge 2 · Town 4 · Factory 4. Three-quarters open ground is deliberate: movement stays fast (§2.3 cost 1), contact comes early, and the match fits Pillar 2’s 10–15-minute envelope — cover is a *purchase*, not the default.

Starting positions (all on Plains; home factory hex left free):

Unit	West	East
Flag Tank	(0,4)	(10,4)
Infantry ×2	(1,3), (1,5)	(9,3) , (9,1)
Artillery	(0,3)	(10,5)
Recon	(0,5)	(10,3)

Guided opening (§2.13.1). `guidedOpening.infantry = (1,5) West / (9,3) East`; `guidedOpening.objective = South (5,7) West / North (6,2) East`. Distinct objectives (T-SCN-07); **5 MP each**, Bridge-free (T-SCN-06); 1 MP of slack against the 6 MP ceiling, in both seats. This map previously named its guided units only in §2.13.1’s lane table; they are declared here now, with the map they belong to.

Non-contention, measured (T-SCN-11; Q22 ruled, Q28 ruled). Q28 rules that “the opposing seat’s Infantry” means **every** Infantry that seat owns, not only its marked guided unit — so this map is checked on **eight** routes, not two. All eight, priced identically: Stub-3 cheapest path, terrain alone with occupancy excluded (Q21), every hex entered counted including the objective, Bridges permitted on opposing routes and forbidden on the two guided lanes.

Infantry	→ North (6,2)	→ South (5,7)
West (1,3)	6 — (2,3)(3,2)(4,2)(4,1)(5,1)B(6,2)	6 — (2,4)(3,4)(3,5)(4,5)(5,6)(5,7)
West (1,5) guided	7 — (2,4)(2,3)(3,2)(3,1)T(4,1)(5,1)B(6,2)	5 — (2,6)(3,6)(3,7)(4,7)(5,7)
East (9,3) guided	5 — (8,3)(7,3)(6,3)w(6,2)	6 — (9,4)F(8,5)(8,6)(7,7)(6,7)(5,7)
East (9,1)	5 — (9,2)(8,2)(7,2)w(6,2)	7 — (9,2)(8,3)(8,4)(7,5)(7,6)T(6,7)(5,7)

Both entries pass, and both report the same pair: 5 against 6. West's South lane costs 5; East's cheapest Infantry route to (5,7) costs 6, from the guided hex (9,3). East's North lane costs 5; West's cheapest route to (6,2) costs 6, from (1,3) over the north Bridge. Strictly longer in both seats, which is the ruled comparison — equality fails, and equality is what this map used to report.

Two structural facts the table makes visible. **Every route to North costs exactly 1 MP more than its hex distance**, because the Water column forces West onto a Bridge and the Woods ring — (6,1)(7,2)(6,3), the only land approaches to (6,2) that are not the Bridge — forces East through cover. **Every route to South is a plain geodesic at 1 MP per hex**, because rows 6–8 are open Plains that the two pass Mountains do not close. That is the map's economy in two lines: the north is priced by terrain, the south is priced by distance alone, and the south is therefore the half where deployment is the *only* thing standing between two Infantry and the same factory.

Why East's second Infantry sits at (9,1) and not (9,5). At (9,5) it was 5 MP from West's South objective — a dead tie with West's own lane, which T-SCN-11 refuses, correctly: a tie is a race and a race is what the invariant exists to prevent. It could not be nudged and stay southern. Every **free** hex in East's southern quarter measures 4–5 MP to (5,7); the only two hexes down there that clear 5 are (10,5) and (10,4) — the Artillery's and the Flag Tank's — so staying southern means pushing a fragile ranged unit or the flag itself onto the outer column for 1 MP of margin. The cause is geometric and worth stating: East's south town (7,6) is only 2 hexes from South (5,7), so by the triangle inequality anything within 3 MP of that town is within 5 MP of that factory. A capturer that covers East's southern town is *necessarily* a racer for West's southern factory.

So the fix is a change of flank, not a shuffle — and it costs East nothing in opening income. East's second Infantry was ever only the early capturer for **one** town; at (9,1) it is adjacent to the north town **(8,1)** and captures on turn 1 exactly as it did at (9,5) for (7,6). One town in the opening either way; north instead of south. What East actually gives up is the southern picket: town (7,6) now waits for a turn-1 Infantry build — 100 of the 200 starting Fame (§2.7), spawning on the deliberately empty home factory (9,4), 3 MP and therefore one turn from (7,6) — and East's opening south flank is Artillery (10,5) behind the Flag Tank (10,4). That is a one-turn, 25-Fame-per-turn delay on one town, paid for out of starting Fame, against a guided lane that stops being a race. It is also the deployment finally agreeing with the sentence this section has always printed: *East's prize is North*.

Terrain as economics, hex by hex. The river spans rows 0–5 only, crossed at two Bridges (–10% defense, §2.3): a unit forcing a crossing under Artillery (range 2–3, no counter, §2.4) buys ground with HP — the intended price. The river deliberately does **not** bisect the map: the southern pass (row 6) is a bridge-free route priced by two Mountains (cost 3, +40%), so it is the *slow* flank, never a free one. The one-hex river also means opposite banks are distance 2 — inside Artillery range, and the prototype ships without line-of-sight blocking (§2.2) — so bank control is contested by fire even before anyone crosses. East's bank carries the Woods band (+20%): cover in exchange for the longer road south. West gets open Plains: speed in exchange for fighting uncovered. The South neutral factory sits between the pass Mountains — +15% defense and a spawn point anchoring the southern flank; the North neutral sits on East's bank, so West must win a bridge to contest it. Rear towns (3,1)/(2,7) west and (8,1)/(7,6) east are the repair points (§2.7): the anti-fortress clause (no repair adjacent to an enemy) keeps the forward ones from healing a frontline garrison.

The tactical question this map asks: *which neutral factory do you race, and which bridge do you contest — knowing the two answers pull a 5-unit army in opposite directions?* West's fast prize is South (open Plains approach); East's is North (no crossing needed). Committing to both splits the army into halves that lose either fight.

Asymmetry and its handicap story. Starting Fame and forces are identical; the asymmetry is purely spatial and intended to be self-balancing. Each seat is closer to a *different* neutral factory — measured, and by exactly **1 MP in each seat**: West's cheapest Infantry route is **5 MP to South against 6 to North**, East's is **5 MP to North against 6 to South** (all four figures in the eight-route table above). Earlier drafts asserted that sentence without pricing it, and for East it was simply false: with an Infantry at (9,5), East was **equidistant** — 5 MP to South and 5 MP to North — so the seat the map called the northern one had no measurable preference at all, and West's southern lane was a tie rather than a lane. The deployment above is what makes the sentence true, and the four numbers, not the sentence, are the claim. Each seat's advantage (West: tempo and open approaches; East: cover and a crossing-free path to its prize) is the other's problem. If §4.1 self-play shows either seat above ~55% win rate, the corrective is the existing §2.9 dial — a per-seat starting-Fame offset — never a terrain rework. This is also the replay lever: the two seats are two different deterministic puzzles (see 2.13.4).

If one side holds both bridges: the other side is not locked out — the southern pass exists precisely so bridge control here is *tempo*, not a topological wall. A double-bridge

holder who then sits earns zero combat Fame and loses the §2.8 cap tiebreak (or draws under the mutual-passivity guard); meanwhile the cross-river Artillery duel and the open southern pass keep combat Fame available to both sides. Full lockout as a map premise is reserved for *The Causeway* (2.13.6), where the tiebreak rules carry the whole anti-turtle load by design.

2.13.3 Match-length reasoning — and the dial it exposes

Estimate: 12–16 turns; cap at 20. Not just the number:

- **Contact turn 3–4.** Homes are 8 columns apart over mostly cost-1 Plains; Tank (move 5) reaches a bridge or the pass mouth in 2–3 turns, Infantry (move 3) in 3–4.
- **Economy online turn 3–4.** Each side’s near Infantry reaches its closest neutral factory in ~2 turns; capture at N=1 (fixed from §2.7’s “start N=1–2” range — Q4, §4.7) flips income the following turn. Income ramps 100 → ~225–250 Fame/turn: a reinforcing unit every 1–2 turns, continuously feeding the fight without flooding it (board space is the throttle, §2.7).
- **Kill speed.** Damage is punchy against 8–20 HP pools (§2.4): units die in 2–3 hits, so bridge and pass fights *resolve* rather than accumulate. Mid-game runs turns 5–12; the flag hunt closes 12–16.
- At ~45 s/player-turn that is ~10–13 minutes — inside §1’s “10–15 typical, ~20 at the cap”.

The match-length dial is factory count × home separation. More neutral factories → steeper income ramp → bloodier, faster mid-game and a wider “objectives held” spread at the cap; fewer → slower armies and more stall risk. Each column of home separation moves the contact turn back roughly one-for-Tank-move. Scenarios tune these two numbers first; unit and terrain stats are never per-map (those belong to rules and balance, §3).

2.13.4 Replayability — configurations, not mechanics

Stratocracy is deterministic end-to-end (Pillar 1, §2.6, §2.9’s tier-invariant AI), so any line that beats the AI once beats it forever; by match ~4 a single mirrored map is a solved puzzle. The replay unit is therefore a **configuration = map × seat × difficulty handicap**, each a distinct deterministic puzzle:

Matches	Configuration	Ships in
1–3	<i>Ferrum Crossing</i> , West seat, Easy → Normal → Hard (§2.9 Fame ladder)	Core
4–6	<i>Ferrum Crossing</i> , East seat , same ladder	Core (seat-select = scenario data + one menu affordance)
7–8	<i>Longwater March</i> , Normal → Hard	Stretch P1 (§4.4 wk 4)
9–10	<i>The Causeway</i> , Normal → Hard	Stretch P2 (wk 4, only after P1)

If no stretch lands, the shipped scope alone moves the cliff from match ~3 to match ~6 — that is what the asymmetric map buys, and why *Ferrum Crossing* is not mirrored. Honest ceiling: determinism means every configuration is eventually solved; this multiplies puzzles, it does not make them unsolvable. The long-term fixes (AI second pass, map-gen MCP toolset) already sit in the GDD’s stretch column and are not re-proposed.

2.13.5 Stretch scenario — *Longwater March* (P1, §4.4 wk 4)

Spec	Value
Dimensions	13 × 8 = 104 hexes — eight rows, not nine, because a 180° rotation only closes on an even row count (§2.13.1)
Starting units per side	5 (standard force — one variable at a time; this map’s variable is factory count)
Factories	6 — one home per side + 4 neutral (<i>above</i> §2.7’s “typical ~4”, <i>inside its</i> “two or more neutral”; <i>pending Q19, §4.7</i>)
Towns	4
Symmetry	180° rotational , $\rho(c, r) = (12-c, 7-r)$ — fair and admittedly dull; chosen so the factory-count dial is the only variable under test. <i>Mirrored</i> was the earlier declaration and is withdrawn: no odd-r rectangle has a vertical mirror axis at any dimension, and the horizontal one exists only on an odd row count — this map is 8 rows (§2.13.1). Rotation costs this map nothing it wanted from mirroring — it is equally distance-preserving, so both seats’ lanes cost 4 MP and the only asymmetry left is the one under test.
Estimated match length	16–20 turns, frequently reaching the cap

r0: m p p p p p T p p p p p m
r1: p p p p F p p p F p p p p
r2: p p p p p p p p p p p p
r3: p F p T p w w p p p p p
r4: p p p p p p w w p T p F p
r5: p p p p p p p p p p p p
r6: p p p p F p p p F p p p p
r7: m p p p p p T p p p p p m

Key coordinates. Home factories: West **(1,3)**, East **(11,4)** — 10 hexes apart, unchanged from the 13 × 9 draft, so §2.13.3’s contact-turn arithmetic is untouched. Neutral factories **(4,1)(8,1)(4,6)(8,6)** · Towns **(6,0)(3,3)(9,4)(6,7)** · central Woods knot **(5,3)(6,3)(6,4)(7,4)** · corner Mountains **(0,0)(12,0)(0,7)(12,7)**.

Every one of those is a ρ -pair under $\rho(c, r) = (12-c, 7-r)$, and the pairing is the map’s whole warrant: (1,3)↔□(11,4) homes, (4,1)↔□(8,6) and (8,1)↔□(4,6) neutrals, (3,3)↔□(9,4) and (6,0)↔□(6,7) towns, (5,3)↔□(7,4) and (6,3)↔□(6,4) woods, (0,0)↔□(12,7) and (12,0)↔□(0,7) mountains. Rows 2 and 5 are all Plains and map to each other. Note that ρ has no fixed hex on an even-row board, so the homes sit on *different rows* (3 and 4) — that is the rotation working, not a drafting slip.

Terrain distribution (104 hexes): Plains 86 · Woods 4 · Mountains 4 · Water 0 · Bridge 0 · Town 4 · Factory 6. Even more open than the shipped map (83% Plains vs. 76%), which is the point: this is the maneuver map, and the only terrain that slows anyone is the four-hex Woods knot standing between the two home rows.

Starting positions (all on Plains; home factory hex left free; East is the exact ρ -image of West, which is what “one variable at a time” means here):

Unit	West	East
Flag Tank	(0,3)	(12,4)
Infantry ×2	(1,2) , (1,4)	(11,5) , (11,3)
Artillery	(0,2)	(12,5)
Recon	(0,4)	(12,3)

Guided opening (§2.13.1): `guidedOpening.infantry` = **(1,2)** West / **(11,5)** East; `guidedOpening.objective` = **(4,1)** West / **(8,6)** East — distinct objectives, 4 MP each, no Bridge on the map at all. 2 MP of slack against the 6 MP ceiling, identical in both seats.

No Water at all — the map that teaches what the chokepoint map can’t: open-field maneuver, Recon (move 7) flanking wide, and expansion tempo deciding who holds 4-of-6 **factories** at the cap. Criterion 2 counts factories *and* captured towns (§2.8), so this map’s denominator is **N = 10** — 6 factories + 4 towns, against N = 8 on *Ferrum Crossing* (§2.11.4). The factory half of that spread is what the extra neutrals buy: a **0–6 factory swing inside a 10-objective sort**, where the shipped map can swing only 0–4.

Match-length reasoning: 4 neutral factories scale income toward 600+ Fame/turn (§2.7) — losses are replaced almost as fast as they land (a Tank kill pays +150 while the victim's economy rebuilds the Tank in under a turn at full income). Attrition drags, the flag hides behind rebuilt lines, and the match leans on the §2.8 tiebreak. That is the point: this is the scenario that *exercises* the combat-Fame tiebreak and the §2.11 scoreboard, which on the shipped map rarely fire.

Tactical question: *how many factories can you hold with the army those factories pay for?* Over-expansion strands Infantry on capture duty while the center Woods fight is lost; under-expansion loses the income race and, at the cap, the objectives-held sort.

2.13.6 Stretch scenario — *The Causeway* (P2, wk 4, only after P1)

Spec	Value
Dimensions	$9 \times 8 = 72$ hexes — eight rows, not nine, because a 180° rotation only closes on an even row count (§2.13.1)
Starting units per side	5 (standard force)
Factories	4 — one home per side + 2 neutral (conforms to §2.7 ~4)
Towns	2
Symmetry	180° rotational , $\rho(c, r) = (8-c, 7-r)$ — fair, and rotation puts each seat's near bridge on the opposite flank (West's is north at (4,2), East's is south at (4,5)), so seat-swap stays non-cosmetic even on a symmetric map. Rotation is not a preference here but the only symmetry available to an odd-r rectangle with an even row count — the vertical mirror exists at no dimension, the horizontal one only on odd H (§2.13.1); the row count is even so that the rotation actually closes.
Estimated match length	8–12 turns

```

r0:  p p p p ~ p p p p
r1:  p T p w ~ p p p p
r2:  p p m F B p p p p
r3:  p F p w ~ p p p p
r4:  p p p p ~ w p F p
r5:  p p p p B F m p p
r6:  p p p p ~ w p T p
r7:  p p p p ~ p p p p

```

Water fills column 4 end to end except Bridges (4,2) and (4,5) — a **full bisection, the deliberate opposite of Ferrum Crossing**. Column 4 is its own p-image, which is why the bisection survives the rotation intact. Homes (1,3)/(7,4) — 6 hexes apart, unchanged. Neutral factories (3,2) and (5,5) each guard the approach to their adjacent bridge from their own seat's bank: +15% defense *and* a spawn point at the chokepoint (§2.7 build-and-spawn makes a held bridge-factory a reinforcement faucet exactly where reinforcements matter). Woods (3,1)(3,3) flank the north bridge on West's bank and (5,4)(5,6) flank the south bridge on East's; each seat's own crossing is the one it defends from cover, and each seat's *far* landing — (5,2) north, (3,5) south — is bare Plains, so attacking a bridge is always the uncovered half of the trade. Single Mountains (2,2)/(6,5) are Artillery perches — range 2–3 covers their bridge hex from +40% cover with no counter (§2.3, §2.4), and reaches the far landing at range 3.

p-pairs, exhaustively: (1,1) $\leftrightarrow$ □(7,6) towns, (3,1) $\leftrightarrow$ □(5,6) and (3,3) $\leftrightarrow$ □(5,4) woods, (2,2) $\leftrightarrow$ □(6,5) mountains, (3,2) $\leftrightarrow$ □(5,5) neutral factories, (1,3) $\leftrightarrow$ □(7,4) homes, (4,2) $\leftrightarrow$ □(4,5) bridges, and the six Water hexes in pairs (4,0) $\leftrightarrow$ □(4,7), (4,1) $\leftrightarrow$ □(4,6), (4,3) $\leftrightarrow$ □(4,4). Everything else is Plains.

Terrain distribution (72 hexes): Plains 52 · Woods 4 · Mountains 2 · Water 6 · Bridge 2 · Town 2 · Factory 4. The tightest board in the set, and the only one where a single terrain type — Water — decides the topology.

Starting positions (all on Plains; home factory hex left free; East is the exact p-image of West). *This map previously specified none, which left §2.13.1's lane figures without an antecedent and made T-SCN-06/07/08 unevaluable against it:*

Unit	West	East
Flag Tank	(0,3)	(8,4)
Infantry ×2	(1,2), (1,4)	(7,5), (7,3)
Artillery	(0,2)	(8,5)
Recon	(0,4)	(8,3)

Guided opening (§2.13.1): `guidedOpening.infantry = (1,2) West / (7,5) East`; `guidedOpening.objective = (3,2) West / (5,5) East`. Distinct objectives (T-SCN-07); 3 MP each, Bridge-free and confined to the seat’s own bank (T-SCN-06); 3 MP of slack. The West lane is (1,2)→(1,1)→(2,1)→(3,2), through the town rather than over the Mountain at (2,2) — the 2-hex route costs 4 MP and the 3-hex route costs 3, which is the case T-SCN-08’s “computes, never infers” wording exists for.

Note what the deployment does *not* do: neither seat starts within reach of a bridge. West’s forward Infantry at (1,2) is 3 hexes from (4,2); the guided opening walks it away from the crossing, to a factory on its own bank. The first lesson on the bisection map is still capture, not crossing — the map earns its lockout premise from turn 3 onward, not turn 1.

Both bridges held — stated in full, because on this map it is the whole design. No naval unit, no other crossing: double-bridge control is a true lockout — the locked-out side cannot reach the enemy flag *or* home factory, so both the flag kill and territorial domination are out of its reach. The rules already make this a trap, and the map exists to prove it: the holder must still *cross* to win decisively; if it sits, it earns zero combat Fame and **loses the cap tiebreak to any opponent who dealt any damage at all** (§2.8 criterion 1); if both sides sit, the mutual-passivity guard calls a draw. Lockout on The Causeway is a tempo platform for a prepared crossing, never a victory condition — the map that demonstrates §1.5 finding #1 (the turtle exploit) is dead.

Match-length reasoning: homes only 6 columns apart, but every route crosses a Bridge; the map forces commitment, and the first successful bridgehead (turn 3–5) usually cascades into the flag kill within 4–6 turns because the defender’s Fame went into defending crossings, not expanding.

Tactical question: *how do you buy a bridgehead when the crossing hex fights at –10% and the far bank fights at +20?* The §2.6 forecast makes the price exact before you pay it — the scenario where the forecast display earns its keep.

2.13.7 Scenario-set summary

Map	Status	Hexes	Units/side	Factories	Towns	Symmetry	Dial it turns
<i>Ferrum Crossing</i>	Ships (§2.10 IN)	11×9	5	4	4	Asymmetric (Fame-correctable)	baseline / seat asymmett
<i>Longwater March</i>	Stretch P1 (wk 4)	13×8	5	6	4	180° rotational	factory count → cap pressure
<i>The Causeway</i>	Stretch P2 (wk 4)	9×8	5	4	2	180° rotational	bridge lockout – decisiven

Neither stretch map may pull work forward of week 4 or block core; if week 4 is consumed by balance (its primary §4.4 purpose), the set stays on paper.

3. AI Architecture — how AI agents are used (roles)

The project is run as a small **agent team directed by a human**. Each role has a clear responsibility, an instrument, and an output that is verifiable.

Role	Owner	Responsibility	Instrument	Verifiable output
Director	Human	Specs, scope line, final balance judgment, review gates, art direction, pipeline writeup	—	This GDD; merge approvals
Systems Engineer	Agent	Author the headless C++ rules (grid, movement, combat, capture, turn loop, win) from spec	Claude Code (headless module, no editor)	Compiling C++ modules
Test Engineer	Agent	Write + run the automation/unit-test suite; block merges on failures	Claude Code + test harness	Passing test suite (the merge gate)
Balance Analyst	Agent	Run headless AI-vs-AI self-play; report win-rate and turn-length; propose stat tuning	Self-play sim harness	Balance logs + tuning diffs
Content / Scenario Designer	Agent	Build maps/scenarios; generate + populate unit/terrain DataTables	Custom MCP toolset in-editor	Scenario assets, data tables
UI Scaffolder	Agent	Generate UMG widget skeletons + bindings for human polish	Claude Code + UE MCP	Wired UMG widgets
Opponent Commander (<i>stretch, runtime</i>)	Agent (LLM)	Play a turn in-product as the enemy	In-game LLM call + move validator	A validated legal move per turn

Pipeline. Claude Code is the agent client. Two surfaces: - **Headless harness** (Systems / Test / Balance roles) — the rules module has **zero engine dependencies**, so these agents compile, test, and self-play in seconds without launching the editor. This is what makes agent authorship fast enough to dominate the build. - **Unreal MCP plugin** (Content / UI roles) — for in-editor operations (scenario building, UMG scaffolding).

Workflow — test-first. Director writes a spec → Test Engineer writes tests against it → Systems Engineer implements until tests pass → Balance Analyst self-plays and proposes tuning → Director reviews and approves. The agent verifies its own work against a spec, not against the Director’s eyeballs.

Worked example — how a role is prompted and constrained. The role table names *what* each agent produces; the constraint is *how*. Every headless system starts as a structured **input spec** the Director hands the Systems Engineer — not a prose ask, but a contract of inputs, an exact formula, and machine-checkable invariants. For combat resolution:

SPEC: Combat resolution (Director → Systems Engineer)
 Inputs: attacker{atk, type, hp, hpMax}, defender{def, type, hp},
 terrainDef% = defender's hex, attackerRange vs distance,
 eff = typeEffectiveness[attacker.type][defender.type] (default
 1.0 everywhere)
 Formula: $\text{dmg} = \max(1, \text{round}(\text{atk} * \text{eff} * (\text{hp}/\text{hpMax}) * (1 - \text{terrainDef\%})) - \text{def})$
 Invariants (Test Engineer asserts each one):
 - terrain defense applies to the DEFENDER's hex only – never the
 attacker's
 - counterattack fires ONLY if the defender survives AND the attacker is
 within the defender's range
 - Artillery (range 2-3) takes zero counter from a range-1 attacker
 - same (attacker, defender, terrain, hp) → identical result
 (determinism; any RNG seeded)
 Determinism: pure function of inputs, no unseeded RNG
 TypeEff: eff ∈ {0.5, 1.0, 1.5}; ships as an all-1.0 table, so RPS
 stays
 positional (§2.4). Populate only if self-play shows the
 triangle
 too weak. eff=1.0 everywhere → existing T-COMBAT-01..08
 unaffected.
 Acceptance: T-COMBAT-01..NN must pass before merge

The Systems Engineer is constrained to that spec; the Test Engineer turns each invariant into a gate *before* implementation exists. This is where the pipeline earns its keep — the gate catches rules the agent hallucinates. A real case:

T-COMBAT-07: Artillery counter-immunity
 Given Infantry attacks Artillery from an adjacent hex (distance 1)
 Then Artillery deals 0 counter damage (its range is 2-3, so it
 can't strike back at 1)
 First agent pass → FAIL: the Systems Engineer generalized "surviving
 units counterattack"
 and wired a range-1 counter for Artillery. Merge blocked → the invariant
 is re-fed →
 re-implemented → T-COMBAT-07 green → merged.

No human eyeballed the diff to catch that; the spec's invariant did, mechanically, at the merge gate. That is the difference between "an agent wrote some C++" and a constrained, self-verifying pipeline.

This crew is the buildable deliverable. The three headless roles above — **Systems Engineer, Test Engineer, Balance Analyst** — are exactly the 3+ coordinating agents instantiated for the standalone agent-crew build: their *game-ready output* is compiling C++ systems, a passing test suite, and self-play balance data for *this* game, handed between them in a fixed order (spec → tests → implementation → balance). The role table's I/O columns are the contract that crew is written against, so the document and the crew stay in lockstep.

Guardrails (non-negotiable). - Perforce checkpoint before any agent-driven edit session; agent permissions treated like production credentials. - The MCP plugin is **experimental, partly undocumented, and runs serially on the game thread** — it is **not on the critical path**; every editor op it does has a manual fallback. - Enable the **AllToolsets** companion plugin, or the MCP server connects but exposes nothing useful. - Claude Code is the documented client for editor-driving (not Claude Desktop).

Provenance ledger — how agent contribution is reported. The deliverable is the game (§1); agent authorship is reported honestly, not as a single hero percentage but as a **per-system ledger** — one row per system, marked by author and verification status, **each Verified row citing the commit and passing test IDs that back it** so the claim is auditable rather than asserted.

Status: live tracker — first rows populated 2026-07-26 from the Assignment-3 agent crew; Repair and Type-effectiveness added and gate-verified 2026-07-29; the rest land as each system is built (wk 1-3, §4.4). Legend: **Author** ∈ {agent, agent+human, human}; **Agent-verified** ∈ {✓, —}; **Evidence** cites the git commit + passing test IDs so any row is independently checkable.

System	Author	Agent-verified?	Evidence (commit · tests)
Combat resolution	agent	✓	Combat.cpp + test_combat.cpp @ 5ffa8d6 · T-COMBAT-01..10 (10/10)
Test suite	agent	✓	test_combat.cpp @ 5ffa8d6 · 17/17 invariants, re-runnable via python run.py
Hex grid & math	<i>pending</i>	—	<i>pending build</i>
Movement & pathfinding	<i>pending</i>	—	<i>pending build</i>
Capture & Fame economy	<i>pending</i>	—	<i>pending build</i>
Turn loop & win / tiebreak	<i>pending</i>	—	<i>pending build</i>
Opponent AI	<i>pending</i>	—	<i>pending build</i>
Data tables (units/terrain)	<i>pending</i>	—	<i>pending build</i>
Repair (owned-tile heal, §2.7)	agent	✓	Combat.cpp::repairAmount @ 5ffa8d6 · T-REPAIR-01..07 (7/7)
Type-effectiveness (§3 spec)	agent	✓	Combat.cpp::effectiveness @ 5ffa8d6 · T-COMBAT-09..10 (neutral, 2/2)
Content / scenario	<i>pending</i>	—	<i>pending build</i>
UI	<i>pending</i>	—	<i>pending build</i>

Four rows are now populated — the headless Combat module built for the Assignment-3 agent crew (github.com/jakemartin/stratocracy-crew, commit [5ffa8d6](#)): **Combat resolution** and its **Test suite**, plus **Repair** and **Type-effectiveness**, all agent-authored from spec/combat_spec.md (+ combat_spec_addendum.md) and verified by the real compile+test gate — a live CrewAI run authored the module and the Test Engineer certified **17/17** on a live g++/clang++ compile+run. (*Commit 5ffa8d6 is published: it is an ancestor of main at [2fcbf32](#) on the public remote, so every link above resolves and the ledger’s “independently checkable” claim is testable by clicking it.*) The remaining rows fill in the same format — commit + passing test IDs — as each system clears the gate.

Rules that keep the ledger honest: - **Scope = the game’s own source only**. The UE engine, the MCP/AIToolsets plugins, and the bought reference template (§4.3, not shipped) are excluded — they aren’t the artifact we built, so they’re not in the denominator. - **Author = human** for anything a human hand-wrote or substantially edited (UI polish, glue). Human work counts as human, so the ledger under-claims rather than inflates. - **Agent-verified = both bars**: (a) covered by agent-authored automated tests that pass in the headless suite (§4.1) **and** (b) signed off at the human review gate (§3). A system is “agent-verified” only if it clears both.

Because the game is deterministic rules + data, this ledger is *checkable*: the test suite is re-runnable and each system’s provenance traces to its commits. It is supporting evidence of the method — the game itself is what ships and is judged.

4. Technical Strategy

4.1 Architecture

- **Headless rules module (C++)** — all of §2 mechanics, no engine deps. Enables sub-second agent test/tune loops and deterministic self-play. Combat damage is a deterministic function of attacker/defender stats, terrain defense, and the attacker’s HP ratio; reachability and routing use Dijkstra/A* over variable terrain cost; any RNG is seeded so tests and self-play stay reproducible.
- **Presentation layer** — Actors + UMG read from the headless state; never own rules.
- **Data-driven** — DataTable/DataAsset for units and terrain; agent-generated, human-reviewed.
- **Test suite** — Unreal Automation tests (or a plain C++ harness on the headless module) covering hex math, pathfinding, combat, capture, and win detection.
- **Balance sim harness** — headless AI-vs-AI self-play, N games → win-rate and turn-length distributions → tuning input.

- **Input:** Enhanced Input. **Save/load:** minimal single-slot for the prototype.

4.2 Engine & tooling

- **Unreal Engine 5.8** (last planned UE5 release; stable base).
- **Unreal MCP plugin** (ships in 5.8, Experimental) + **AllToolsets** — editor operations for the Content/UI agent roles. Serial-on-game-thread; off the critical path.
- **Custom MCP toolset** — `place_terrain`, `place_unit`, `set_objective`, `validate_scenario`, authored via `ToolsetDefinition` (Python or C++) so the Content agent builds/edits scenarios in-editor.
- **Reference implementation** — cross-check the agent's grid/pathfinding against a mature template (e.g. *Advanced Turn Based Tile Toolkit*). **License caveat: verify AI-usage terms before an agent reads template code.**

4.3 Build approach

Author the **core in C++ with the agent** (the on-thesis path — hex math, pathfinding, and combat are where agents are strongest and least likely to hallucinate). Keep a bought template installed **only as a reference to check output against**, not as the shipped codebase, so the “agents built the game” claim holds.

4.4 Milestones (7 weeks)

Wk	Goal
1	Headless C++ core — §4.11 rows 1–3 (grid and hex math, the §4.8 tables, movement and pathfinding) on top of the already-verified Combat/Repair at 5ffa8d6 — + test suite. Playable via debug commands. Win and tiebreak detection is row 5, bundled with the turn loop and landing wk 3 (Q23); this cell used to name it here, one week ahead of the Capture & Fame row it depends on. UI-scaffolder agent starts UMG widget skeletons in parallel — the whole game is played through UI, so it can't wait until wk 2. Engine presentation + UI wiring (select/move/attack) onto the wk-1 skeletons, plus the one scenario loading, validating and rendering, and the §4.10 save/replay format + headless replayer (Q20, ruled). The format is a <i>test instrument</i>, not a feature: T-INT-02's input file is a save, so the week-2 parity gate cannot run without it — and neither can the week-4 self-play logs (T-SAVE-07). Week 2's command set is exactly {Move, Attack} (§4.9): Capture, Build and EndTurn arrive with §4.11 rows 4–5 in wk 3, so a wk-2 log's entries are all {turn 1, one side}. The wk-2 parity and replay gates therefore run over that subset and re-open when it widens — T-INT-01/04 and T-SAVE-04 close here, the rest do not (§4.11 rows 9–10). That is a real gate rather than a placeholder: T-INT-02 replays a <i>fixed</i> log, so its divergence surface is the state-mutating math only, and every field of the §4.10 hash is an integer except the transients inside <code>resolveDamage</code> — which Attack already reaches. What it cannot cover is code that does not exist yet, which is exactly why rows 4–5 re-open it. No save button, slot, or <code>USaveGame</code> wrapper here; those are week 5. Move + attack only — no capture, no production, no AI opponent. Capture + production (§4.11 rows 4–5), then the baseline objective-seeker AI (row 0, <i>making partial solution</i>, Rows 4–5
2	

Wk	Goal
3	0) → working vertical slice . Rows 4–5 add precisely the three commands week 2 lacked, so the wk-2 gates re-run over the complete set and close here : T-INT-02/03/05 and T-SAVE-01/03/05/06 on rows 4–5, and T-SAVE-02 on row 6, whose determinism gate (T-AI-06) it composes. Only T-SAVE-07 still waits — for wk 4’s self-play logs. AI second pass (utility + threat map) and the custom MCP scenario toolset follow only if the slice lands early; both are off the critical path (§4.2, §4.11). Self-play balance sims and tuning — every match emitted in the wk-2 §4.10 format, so T-SAVE-07 (harness compatibility) closes here , not in wk 5: one format, no dialect drift between a save and a balance log; scenario polish (additional scenarios only as stretch).
4	UI polish, feedback/juice, the save-slot UI and its slot I/O — the §4.10 format and headless replayer landed in wk 2 (Q20), so what remains here is the player-facing surface only: the single <code>slot0</code> file, the <code>USaveGame</code> wrapper, and whatever §2.11 decides about overwrite-confirm (§4.10 records that surface as unowned); onboarding.
5	Playtest, bug fix, balance lock. LLM-commander stretch only if on track .
6	Final polish, package/build the shippable game (the deliverable), and write up the agent pipeline + provenance ledger (supporting evidence).
7	

On weeks 2–3 (Q23, ruled). The vertical slice sits in week 3, not week 2, because §4.11’s critical path is 1 → 3 → 4 → 5 → 6/8: the baseline AI (row 6) needs the turn loop (row 5), which needs Capture & Fame (row 4). A week-2 slice *with* an AI opponent would have required building rows 4–6 a week before §2.10 says capture and production land — the document previously asserted both and could not have delivered either. Week 2 therefore delivers move + attack against a static board, which is the honest subset of the slice that rows 1–3 actually support, and §2.10’s “*these land wk 3, not wk 1–2*” now describes the schedule rather than contradicting it. **The save/replay half of the same question (Q20) is now ruled — and split rather than moved.** The §4.10 *format and headless replayer* go to **week 2**, where T-INT-02 needs a save file as its input, and are therefore in hand by **week 4** for the self-play logs T-SAVE-07 validates; only the **save-slot UI and slot I/O** stay in week 5. The two rulings run on one principle in opposite directions: each piece lands in the week the thing that consumes it runs, so a milestone that outran its dependencies moved *later* (Q23) and an instrument its own gates outran moved *earlier* (Q20). Stated once, so the table stops drifting: **a format is a test instrument; slot I/O is a feature**. One consequence of that principle needed stating too, because the first application of it promised a week-2 gate this document’s own build order could not support: **an integration or replay gate is scoped to the command set of the log it runs on**. It *runs* as soon as that log can be produced and it *closes* only when the log carries every §4.9 command — so week 2’s {Move, Attack} pass is a run, not a closure, and §4.11 rows 9–10 now state those two dependency sets separately instead of one. The seam that repairs: row 9 required “rows 1–5 built” while Q23 had just limited week 2 to rows 1–3, so §4.4 and §4.11 described two schedules for the third time. Neither week number moved to fix it; the gate’s scope was named. And because a gate that runs green over a subset is not a verified system, **no §3 ledger row flips on a partial pass** (Q29).

4.5 Risks & mitigations

Risk	Mitigation
Scope creep (16 scenarios is a commercial game)	Hard MVP line; 1 scenario + flag win is complete
UI underestimated	Treat as core; start week 2
AI rabbit hole	Ship the heuristic; LLM behind a toggle
MCP plugin experimental/flaky	Off the critical path; Perforce checkpoints; manual fallback
Pacing (the original's flaw)	Small maps, decisive damage, turn cap + tiebreak
Agent code quality	Test-first; human review gates; headless module keeps loops fast

4.6 Token budget

AI-agent work is the project’s main variable cost, so it is budgeted explicitly. Two surfaces consume tokens: **development-time authoring** (agents writing, testing, and tuning the game across the 7 weeks) and the **runtime LLM commander** (a stretch feature, billed only if it ships). Rates below are Anthropic’s official published API prices, **verified against platform.claude.com on 23 Jul 2026** — Sonnet 5 introductory (\$2/\$10 per M in/out, \$0.20/M cache read), Opus 4.8 (\$5/\$25, \$0.50/M cache read), Haiku 4.5 (\$1/\$5, \$0.10/M cache read). *Note: the Sonnet 5 introductory rate holds **through 31 Aug 2026**, then reverts to the \$3/\$15 standard rate; the 7-week jam (Jul–late Aug 2026) falls entirely inside the introductory window, so these figures apply for the whole project.* As a conservative buffer, estimates carry a **~30% token-overhead margin** on top of raw text length (accounting for tokenizer overhead, tool-call framing, and re-reads). **Prompt caching** (cache reads billed at 0.1× the input rate) is assumed for the stable spec/rules context the agents re-read on every task.

A. Development-time authoring — the bulk. Workhorse model **Claude Sonnet 5** (\$2 / \$10 per M input/output, \$0.20/M cache read), with occasional **Opus 4.8** escalation (\$5 / \$25 per M) for hard reasoning. The unit of work is one substantial agent task — author or iterate a system together with its tests — spanning many tool calls and file reads.

Line item	Estimate
Tokens per substantial task	~300k input (~⅔ from cache: ~100k fresh + ~200k cache read) + ~45k output = 345k . The ~30% overhead margin above is already inside these figures and is not applied again below
Cost per task — Sonnet 5	100k × \$2/M + 200k × \$0.20/M + 45k × \$10/M = \$0.20 + \$0.04 + \$0.45 = \$0.69
Cost per task — Opus 4.8	100k × \$5/M + 200k × \$0.50/M + 45k × \$25/M = \$0.50 + \$0.10 + \$1.125 = \$1.725 (quoted to the cent as \$1.73; \$1.725 is the exact figure , and every escalation delta in this section is computed from it, not from the rounded display)
Task volume	5 agent roles × ~6 tasks/wk × 7 wk ≈ 210 tasks
Sonnet-only base	210 × \$0.69 ≈ \$145 · 210 × 345k ≈ 72M tokens
Opus escalation — a line, not a multiplier	~15% of tasks — 15% of 210 = 31.5, i.e. 32 to the nearest whole task — run on Opus <i>instead of</i> Sonnet, same tokens at a higher rate. The delta is taken unrounded : \$1.725 – \$0.690 = \$1.035/task . So 32 × \$1.035 = \$33.12 ≈ +\$33 . \$1.035 is the only escalation delta used anywhere in §4.6
Dev-time subtotal	≈ 72Mtokens · ≈ \$178 — unrounded, \$144.90 + \$33.12 = \$178.02 ; rounding first (\$145 + \$33) reaches the same \$178, so the printed figure is stable either way. Downstream lines scale the unrounded per-task costs, never this rounded total

The escalation is priced once. It is a *delta* between two models on the same task, so it belongs inside the subtotal and there is no further uplift after it — an earlier draft folded it into the subtotal silently and then applied it a second time to reach \$225, which is why

three figures in one table disagreed. Every number above is re-derivable from the two rate lines and the task count, which is the property that made the error visible. The escalation assumes Opus **substitutes** for Sonnet on those tasks; if it instead runs *in addition* (a re-run rather than a substitution) the line is $32 \times \$1.725 = \$55.20 \approx +\$55$ and the subtotal $\$144.90 + \$55.20 = \$200.10 \approx \200 .

B. Runtime LLM commander — stretch, per match. Model **Claude Haiku 4.5** (\$1 / \$5 per M) for turn latency; Sonnet 5 as a quality upgrade. Each AI turn serialises the board and its legal moves, asks the model to rank one, then validates and applies it (§2.9).

Line item	Estimate
Tokens per AI turn	~2.5k fresh input + ~1.5k cached rules + ~0.4k output = 4.4k
Cost per AI turn	$2.5k \times \$1/M + 1.5k \times \$0.10/M + 0.4k \times \$5/M \approx$ \$0.0047 (Haiku 4.5)
Per match (~20 AI turns)	88k tokens · $\approx$ \$0.09
200 self-play + playtest matches	$\approx$ 17.6M tokens · $\approx$ \$19 (Haiku 4.5)
Same volume on Sonnet 5	$\approx$ \$37 at the introductory \$2/\$10 rate this project runs inside · $\approx$ \$56 at the \$3/\$15 standard rate that resumes 1 Sep 2026

Headline. Development authoring dominates. Dev-time alone is $\approx$ **72M tokens** · $\approx$ **\$178**; the stretch runtime commander adds $\approx$ **18M tokens** and **\$19 (Haiku 4.5)** to **\$37 (Sonnet 5)**. The whole jam therefore lands at $\approx$ **90M tokens** and $\approx$ **\$178 without the commander, $\approx$ \$197–\$215 with it.** Cost scales linearly with task volume, and the overrun case is stated rather than buried in a wide band. At **1.5× task volume** — 315 tasks rather than 210, of which 15% escalate to Opus (47.25, i.e. **47** to the nearest whole task) — the dev-time line is $315 \times \$0.69 + 47 \times \$1.035 = \$217.35 + \$48.645 =$ **\$265.995**, i.e. $\approx$ **\$266**. The overrun is in *agent task volume*; it does not scale the 200-match playtest plan, so part B carries across unchanged and the all-in overrun is **\$266 + \$19 $\approx$ \$285** with the Haiku 4.5 commander and **\$266 + \$37 $\approx$ \$303** with Sonnet 5. **\$303 is the ceiling.** Each of those is a derivation rather than a round number — as is every figure in the two tables above. (An earlier draft printed **\$267** here. That was $1.5 \times$ the *rounded* \$178 subtotal, not a re-derivation from the per-task lines, and it sat \$1.005 above what its own stated inputs produced ($\$267 - \265.995 ; the often-quoted \$1.24 came from the retired \$1.03 delta) — the second arithmetic fault this table has surfaced by being fully re-derivable, and the reason the escalation delta is now quoted unrounded at \$1.035 throughout.)

4.7 Pending-system gate plan — the eight **pending** ledger rows

Each stub below follows the proven Combat-spec shape (§3): Inputs, Formula or state transition, Invariants (one per assertable rule), Determinism, Acceptance test IDs. All rules code is **headless** — namespace `strat`, pure C++17, zero engine dependencies, compiled by the same `g++/clang++ + python run.py` gate that certified Combat (§3 ledger). Where a stub needs a rule the GDD does not state, the gate is parameterized on a numbered open question — the Director rules, the gate then pins the ruling. The register below is the single place their extent is stated; nothing else in this document names a range of them, because every range named elsewhere has gone stale within a revision.

Shared conventions (Director-owned contract):

- **Coordinates:** axial (*q*, *r*), pointy-top hexes (§2.2). Distance is the standard axial hex metric $(|dq| + |dr| + |dq + dr|) / 2$ — pure integer math. **Two conventions coexist deliberately, and the conversion between them is part of this contract:** §2.13’s authored maps use odd-r offset (*col*, *row*), which is what a human reads off an ASCII grid, while the module uses axial because the distance metric above is only clean in axial. The scenario loader (Stub 7) converts odd-r → axial on import — $q = col - (row - (row \& 1)) / 2, r = row$ — and that conversion is itself gated (T-SCN-05). Neither convention leaks into the other’s layer: no authored file stores axial, and no module code stores (*col*, *row*).
- **Canonical hex order** (the subject of T-HEX-07): a total order over hexes — **ascending *r*, then ascending *q*.** Used everywhere enumeration order could leak into behavior or bytes: reachable-set enumeration, movement/AI tie-breaks (T-MOVE-04, T-AI-06), scenario serialization and hashing (Stub 7), and the §4.10 canonical state hash.
- **Numbering:** stubs 1–8 below are also the build-order row numbers (Build order table).

SPEC STUB 1: Hex grid & math (Director → Systems Engineer)
 Inputs: map dimensions (Q1); axial coords (q, r).
 Functions: neighbors(hex) – the six adjacent hexes in a fixed, documented enumeration order, out-of-bounds candidates filtered; distance(a, b); inBounds(hex).
 Invariants:
 T-HEX-01 every in-bounds hex has exactly six neighbor candidates in the fixed order; filtering removes only out-of-bounds hexes (§2.2 "six equal neighbours")
 T-HEX-02 distance is a metric: $d(a,a)=0$; $d(a,b)=d(b,a)$; triangle inequality
 T-HEX-03 $d(a,b)=1 \iff b \in \text{neighbors}(a)$
 T-HEX-04 direction fairness: each of the six unit steps has distance exactly 1 – no direction is cheap the way diagonals are on squares (§2.2)
 T-HEX-05 inBounds agrees with the Q1 dimensions; every hex reference the engine or a scenario hands the module is bounds-checked, never trusted
 T-HEX-06 single distance definition: combat's range checks (T-COMBAT-06..08 @ 5ffa8d6) consume THIS distance function – Artillery at distance 1 cannot counter, per the verified module, with no second metric to drift
 T-HEX-07 canonical order + determinism: sorting any hex set by (r asc, q asc) is total, stable, and platform-independent; neighbors() enumeration order is fixed across runs and compilers
 Determinism: pure integer functions; no state.
 Acceptance: T-HEX-01..07.

SPEC STUB 2: Data tables (units/terrain) (Director → Systems Engineer)
 Defined in full in §4.8 (schemas, both-side structs, parity gate).
 Invariants
 T-DATA-01..06 are specified there once – this stub defers to §4.8 rather than duplicate a contract that must never fork.

SPEC STUB 3: Movement & pathfinding (Director → Systems Engineer)
 Inputs: unit {move, moveClass}; terrain move costs (§2.3 via Stub 2); current occupancy; start hex.
 Transition: reachable set + cheapest paths via Dijkstra over terrain cost (§4.1); executing a move relocates the unit along the chosen path (§2.5).
 Invariants:
 T-MOVE-01 reachable set is exact: a hex is in the set $\iff$ its cheapest path cost $\leq$ Move – "the real move set, not an estimate" (§2.5)
 T-MOVE-02 costs per §2.3: Plains 1, Woods 2, Mountains 3, Town/Bridge/Factory 1; Water impassable to land – a land path across a Water span exists $\iff$ it crosses on Bridge hexes (§2.3, "the only hex a land unit crosses Water")
 T-MOVE-03 occupancy: a move never ends on an occupied hex (§2.5, one unit per hex). Pass-through of friendly-occupied hexes: parameterized on the Q3 ruling; until ruled, the gate asserts the conservative reading (occupied hexes block pathing entirely)
 T-MOVE-04 the executed path is minimal-cost; ties between equal-cost paths are broken by canonical hex order, so the route is reproducible
 T-MOVE-05 no zones of control: moving adjacent to an enemy costs nothing extra and freezes nothing (§2.5 – ZOC is cut)
 T-MOVE-06 determinism: same state $\rightarrow$ identical reachable set and identical path
 (T-MOVE-07 reserved: Recon's "ignores some terrain cost" (§2.4) – blocked on the Q2 movement-class ruling; no gate is written until the rule exists.)
 Determinism: pure; all tie-breaks canonical.
 Acceptance: T-MOVE-01..06 (07 reserved on Q2).

SPEC STUB 4: Capture & Fame economy (Director → Systems Engineer)

Inputs: game state; commands Build{factoryHex, unitId}, Capture{unit};
the §2.7

income/award values; §2.4 costs via Stub 2.

Transition: income accrual; build-and-spawn; capture progress; kill awards.

Invariants:

T-FAME-01 single pool (§2.7): income, kill awards, and spending all mutate one

per-side fameTotal; combat awards ALSO accrue a separate fameCombat

counter (the §2.8 tiebreak criterion-1 sort key); passive income

never touches fameCombat

T-FAME-02 income: each held factory pays +100/turn, each held town +25/turn

(§2.7); accrual timing per the Q8 ruling

T-FAME-03 build: deducts the exact §2.4 cost (Infantry 100, Recon 150, Artillery 200, Tank 300); refused if unaffordable; fameTotal is

never negative

T-FAME-04 spawn: on the factory hex if free, else an adjacent free hex, else

the build waits (§2.7); waiting-build semantics per Q8

T-FAME-05 capture: Infantry only (§2.7, §2.4); completes after N turns of

holding (N per Q4); interruption/reset semantics per Q4

T-FAME-06 a captured objective's income flips to the new owner (§2.7)

T-FAME-07 kill awards: ~half the victim's Fame cost, small undamaged-strike

bonus, flag kill +500 and the match ends (§2.7) – exact values per

Q5/Q6; the gate pins whatever the Director rules

T-FAME-08 no Fame cap: fameTotal is unbounded; deployment is throttled by

board space only (§2.7)

T-FAME-09 determinism: same state + command → identical Fame deltas and state

Determinism: pure state transitions; no RNG anywhere in the economy.

Acceptance: T-FAME-01..09.

SPEC STUB 5: Turn loop & win / tiebreak (Director → Systems Engineer)
Inputs: game state; per-unit act flags; the turn counter and cap (Q7, stored in the scenario file, Stub 7); commands incl. EndTurn{}.
Transition: I-GO-U-GO alternation (§2.1); win/loss/draw evaluation (§2.8); start-of-turn repair application (§2.7).
Invariants:
T-TURN-01 strict alternation; each unit acts at most once per own turn, in any order the owner chooses (§2.1)
T-TURN-02 flag death ends the match immediately – Decisive win for the killer, loss for the owner (§2.8)
T-TURN-03 territorial domination: controlling every factory on the map at the start of your turn ends the match immediately, ranked Decisive (§2.8; factories only, towns excluded)
T-TURN-04 at the turn cap, the attrition tiebreak resolves in the exact §2.8
T-TURN-05 order: combat Fame → objectives held → surviving HP → draw mutual-passivity guard: both sides' fameCombat == 0 at the cap → immediate draw, with NO fall-through to objectives held (§2.8)
T-TURN-06 criterion 2 (objectives held, X of N) is evaluated only when both sides fought and their fameCombat is equal (§2.8)
T-TURN-07 result tiers are categorical: Decisive > Marginal > Draw, regardless of Fame totals – Fame is only the sort key inside criterion 1 (§2.8)
T-TURN-08 repair fires at the start of the unit's turn exactly when the verified repairAmount says so (owned Town/Factory, no adjacent enemy, +25% max HP floored, min 1, capped – T-REPAIR-01..07 @ 5ffa8d6); this gate asserts the turn loop calls it at the right moment with the right board facts, nothing more
T-TURN-09 determinism: the same command sequence from the same scenario → identical result tier and identical state at every step
Determinism: pure state machine; the §4.10 hash is taken from this state.
Acceptance: T-TURN-01..09.

SPEC STUB 6: Opponent AI (baseline) (Director → Systems Engineer)
Inputs: full game state (the AI cheats at nothing and sees only real state); the §2.9 baseline routine; default buildlist (§2.9).
Transition: economy phase then unit phase, emitting ordinary commands that the rules module validates like any player's (§2.9).
Invariants:
T-AI-01 legality: every AI command passes the same validation as a player command; zero rejected commands across N self-play games
T-AI-02 economy phase: at each held factory, if a buildlist unit is affordable, one is built – the AI spends and replaces losses rather than hoarding (§2.9)
T-AI-03 capture behavior: an idle Infantry near an uncaptured, undefended factory/town moves onto it to capture (§2.9) – objectives stay live on both sides
T-AI-04 attack preference: the enemy flag if in reach, else the best expected-damage target; Artillery fires from maximum standoff (§2.9)
T-AI-05 strictly-losing-attack guard: the AI never makes an attack in which its unit dies and trades down (§2.9)
T-AI-06 determinism: same state → same move; every scoring tie is broken by a stated deterministic rule (canonical hex order for position ties; remaining tie dimensions per the Q9 ruling)
Determinism: pure function of state; difficulty changes only starting Fame (§2.9), never the routine.
Acceptance: T-AI-01..06, plus a self-play smoke run: N headless AI-vs-AI games all terminate at or before the cap with a valid result tier.

SPEC STUB 7: Scenario file & validator (Director → Content/Scenario Designer,
Systems Engineer for the

loader)
 Format: one versioned JSON file per scenario; strat::loadScenario parses and
 validates it headless. Fields:
 formatVersion int unknown → refuse load
 scenarioId string stable identifier
 scenarioHash string hash of the canonical serialization
 (fields
 order).
 FIELDS
 field can
 discipline
 is not
 after
 sits here.
 map object width/height (Q1) + per-hex terrain Id
 (row-major in canonical hex order; Ids
 are
 ownership array §2.3 / Stub 2 row names)
 a home
 factories/towns
 placements array {side, unitId, hex, isFlag} – starting
 units;
 the
 startingFame object flag is "a designated Tank")
 setup
 scenario field
 turnCap int the §2.8 cap (value per Q7)
 guidedOpening array one entry per side, {side, infantry,
 objective} – that seat's opening-
 capture lane
 are HEX
 seat's
 the
 because
 ID
 serialize in
 not
 content-only.
 declares
 every
 no
 layer
 scenario
 presentation
 of
 symmetry enum REQUIRED. `rot180` or `none`, the two
 values
 is

in this order, hexes in canonical hex
 The order is load-bearing, so NEW
 APPEND AT THE TAIL of this list: an
 append-only preimage means adding a
 never reorder an existing one (the
 that appended `type` last in the combat
 addendum, Part A). Serialization order
 validation order – a field is validated
 whatever it references, wherever it
 width/height (Q1) + per-hex terrain Id
 (row-major in canonical hex order; Ids
 §2.3 / Stub 2 row names)
 initial owner of each capturable hex –
 factory per side, neutral
 unowned (§2.7)
 {side, unitId, hex, isFlag} – starting
 isFlag is valid only on a Tank (§2.4:
 flag is "a designated Tank")
 per side; 200/200 baseline (§2.7). The
 difficulty handicap (§2.9) is a match-
 parameter applied on top, not a
 the §2.8 cap (value per Q7)
 one entry per side, {side, infantry,
 objective} – that seat's opening-
 (§2.13.1). `infantry` and `objective`
 references: the deployment hex of the
 marked Infantry (§2.11.6 turn 1a) and
 neutral Factory hex it walks to. A hex
 identifies the placement uniquely
 T-SCN-02 already forbids two placements
 sharing one, so this adds no instance-
 concept to the schema. Entries
 the module's side enumeration order,
 authoring order, so the hash is
 Required on every scenario: §2.13.1
 the reachability invariant shared by
 map, and §2.11.6's turn-2 directive has
 other source for the pair. The guidance
 reads this field from the loaded
 directly – marked/locked is
 state, not rules state, so it stays out
 the Stub-8 snapshot.
 REQUIRED. `rot180` or `none`, the two
 §2.13.1 declares (Q24, ruled). The enum

narrow by SCOPE, not by impossibility,
 and the difference is load-bearing. An odd-
 r rectangle has no VERTICAL mirror axis
 at any dimension (§2.13.1) – but on an ODD row
 count it does have a HORIZONTAL one,
 is $\mu(c, r) = (c, H-1-r)$: with H odd, H-1
 every even, so r and H-1-r share a parity,
 carries row keeps its offset and the column is
 declares unchanged. 11 x 9 Ferrum Crossing
 drawn to that axis geometrically and still
 gives `none`, because its terrain is not
 different it. So `mirror` is a claim the schema
 is an author no way to STATE, not one the
 validator could never ACCEPT – two
 failure modes, and Q26 ruled which one
 is intended: the narrow enum stands.
 Admitting the value could widen nothing: rot180 needs an even H and a
 horizontal mirror an odd one, their
 mirror composition would be the vertical
 `none` that never exists, so at most ONE non-
 value is well-formed on any given map.
 not Absent or unrecognized is a hard load
 failure, never a default of `none` – a
 is scenario that forgets to declare must
 the silently claim the weakest claim. This
 the not a balance or layout field: it is
 hex by AUTHORED CLAIM that T-SCN-09 verifies
 because hex, and it exists in the schema
 Appended at §2.13.1 asks validate_scenario to
 above; machine-verify a declaration the file
 costs previously had no place to make.
 and is the tail per the scenarioHash policy
 Invariants: it moves every scenario's hash, which
 T-SCN-01 exactly one isFlag placement per side, and it is a Tank; the nothing while no scenario file exists
 flag appears in no buildlist anywhere – "not producible" (§2.4) is the point of the policy once one does.
 a scenario/production fact, not a fifth unit row
 T-SCN-02 structural validity: every hex reference is in bounds (T-HEX-05);
 every terrain and unit Id resolves to a Stub-2 row; no two
 placements share a hex (§2.5)
 T-SCN-03 economy validity: each side owns exactly one home factory at
 start, and at least two neutral factories exist in contested ground
 (§2.7, "~4 factories total")
 T-SCN-04 playability: the two flags are mutually reachable by land
 movement (Stub 3 pathing, Bridge rules respected) – a scenario cannot
 be born stalemated
 T-SCN-05 coordinate conversion (Shared conventions): odd-r (col, row) →
 axial (q, r) round-trips for every in-bounds hex of the declared
 dimensions, and the loaded map's adjacency matches the
 authored grid's: no authored file stores axial. and no loaded state

stores (col, row)

T-SCN-06 opening-capture lane (§2.13.1): for EACH guidedOpening entry,
a land path exists from `infantry` to `objective` costing
≤ 2 x Move of the capturing unit row – T-DATA-03's single
CanCapture row, §2.4 Infantry Move 3, so 6 MP – and crossing

NO Bridge hex. The ceiling is DERIVED from the loaded table,
never a literal: a §2.4 Move change re-prices the gate instead of
silently passing it, and the capturing row is found by CanCapture so
§2.7's Infantry-only rule and this lane can never name different
units. Cost counts every hex entered including the objective itself
(Factory MoveCost 1, §2.3) – the same accounting as T-MOVE-01,
so the validator and the reach highlight cannot price one lane
two ways. Excluding Bridge (and Water being land-impassable, §2.3)
confines the lane to the seat's own bank, which is what makes
the first lesson uncontested rather than a crossing. Asserting the
an existential over the NAMED hex is deliberate. §2.13.1 states
EXISTENTIAL ("at least one Infantry deployment hex must have a
land path...") and, alongside it, a NAMING requirement (the
scenario file names that unit, so §2.11.6's turn-1a marked
Infantry is the one already standing on the lane); quantifying
over the named hex collapses both into a single check. Written
to the other way round – find ANY qualifying hex, then compare it
lane the named one – the validator passes a map whose qualifying
belongs to a unit turn 1a never marks: the reachability rule
satisfied to the letter while the marked Infantry stands
somewhere else, and the guided opening's first directive
walking the player down a lane nobody priced.
The gate asserts ARRIVAL ONLY: the turn the tile flips is
N-dependent (Q4) and is asserted nowhere here, matching
§2.13.1's "capturing by turn 2, never captured by turn 2."

T-SCN-07 opening-capture naming (structural; no pathing, so this half
of the check lands with rows 1-2): exactly one guidedOpening entry
per side; `infantry` is the hex of a starting placement of THAT side
whose unitId is the CanCapture row and whose isFlag is false;
`objective` is a Factory hex that `ownership` leaves neutral (§2.7); and
the two entries name DIFFERENT objectives – §2.13.1's "the seat's own
neutral," gated at its distinctness floor. Distinctness is now
exactly that: a FLOOR, not the requirement. The stronger
non-contention property §2.13.1 claims is RULED (Q22) and is
asserted by T-SCN-11, which prices a path and therefore lands
with row 3 rather than with this check. Two seats can name
DIFFERENT objectives and still race each other to one of them
– which is what T-SCN-11 exists to refuse and what this
invariant, being structural, cannot see.

T-SCN-08 measured, not inferred: the validator COMPUTES and REPORTS
each entry's lane cost as an integer, from Stub-3 pathing. The
declared-symmetry flag (§2.13.1) is not an input and cannot substitute.
Not because the flag is unreliable in principle – a verified
rot180 IS an isometry and does imply equal cost between IMAGE lanes
(§2.13.1, T-SCN-09) – but for three reasons that hold even when it is
right: (i) it is an AUTHORED DECLARATION, and measurement is the only
is thing that catches the author declaring the wrong one, which
and what happened here (the flag said mirrored; the numbers said 3
4: the numbers were right. §2.13.1): (ii) a `none`-declared

map offers nothing to infer from at all, and `none` is what the SHIPPED map declares (§2.13.2); (iii) even a verified rot180 forces equal lane cost only if the two guidedOpening entries are themselves rho-images, which §2.13.1 does not require and T-SCN-09 does not assert (Q25) – so equal numbers on a symmetric map are a result, not a theorem. And no symmetry argument of any kind prices a lane, because cheapest path is not fewest hexes.

Fixtures:

- (a) The Causeway (§2.13.6) PASSES, reporting 3 and 3. Its West lane (1,2)->(1,1)->(2,1)->(3,2) is THREE hexes costing 3 MP through the town; the TWO-hex route over the Mountain at (2,2) costs 4 (§2.3). An implementation that counts hexes instead of summing MoveCost reports 2, and nothing else in this suite catches it.
- (b) Longwater March (§2.13.5), rot180 on 13 x 8, PASSES and reports 4 and 4 – COMPUTED equal, never assumed equal.
- (c) A scenario whose lanes both cost 7 FAILS the T-SCN-06 ceiling, and the refusal reason CARRIES BOTH MEASURED INTEGERS: an author needs to read the measured 7 against the 6 MP CEILING, not merely "too far" (Determinism, "refuses with a reason"). This is the only "against" in this stub that is not owning-vs-opposing: it compares a MEASUREMENT to a BUDGET, and it SAYS SO at the site – the right-hand term is printed as "the 6 MP ceiling", never as a bare integer, per T-SCN-11's print convention. NAMING THE RELATION IS THE WHOLE OF THE DISAMBIGUATION. Integer order does not separate the two, because a T-SCN-11 refusal whose opposing route is cheaper prints the larger integer first as well; the earlier reading – that this was the one relation whose failing form led with the larger number – was false and is withdrawn.

The reported integers are the source of truth for §2.13.1's lane table, so a map edit that lengthens a lane surfaces as a changed number rather than a still-green boolean.

T-SCN-09 §2.13.1's declared symmetry is VERIFIED, not trusted – the gate "machine-verified in axial" clause names, and which no other invariant asserts (T-SCN-05 supplies the axial frame and asserts nothing about symmetry).

formed. `symmetry` == `none` asserts nothing and is always well-formed. `symmetry` == `rot180` asserts, for EVERY in-bounds hex h:

- terrain(rho(h)) == terrain(h);
- ownership maps onto itself with the two sides EXCHANGED –

each home factory's image is the other side's home, each neutral capturable's image is neutral (§2.7);

- the placement set maps onto itself with sides exchanged

and unitId and isFlag preserved, which is what §2.13.5 and §2.13.6 mean by "East is the exact rho-image of West."

SCOPE per Q25: guidedOpening is NOT bound, so a rot180 map may name lanes that are not each other's image and report unequal costs (T-SCN-08 (iii)). Both stretch maps satisfy the stronger reading as drawn, so ruling Q25 either way moves no layout.

loaded rho RUNS IN AXIAL, after the T-SCN-05 conversion, because no state holds (col, row) to rotate. On an even row count §2.13's authored rho(c, r) = (W-1-c, H-1-r) and the 180-degree isometry are the same map, and in axial it is one parity-free affine map:

$$\text{rho}(q, r) = (W - H/2 - q, H - 1 - r)$$

The EVEN ROW COUNT is a PRECONDITION, not a comparison: on odd H that constant W - H/2 is a half-integer, so no hex has a hex image

and the file is REFUSED WITH A REASON before any comparison runs

(§2.13.1; Q24, ruled). On 9 x 9 the constant is 4.5 and (1,1) rotates to column 6.5 – one refusal, rather than the offset

index permutation quietly producing geometrically meaningless comparisons.

Structural: no pathing, so this lands with rows 1-2 (§4.11).

(T-SCN-10 reserved: verification of a HORIZONTAL mirror declaration on an odd row count. In axial, after the T-SCN-05 conversion, that map is

$$\mu(q, r) = (q + r - (H-1)/2, H-1-r)$$

with the ODD row count as the precondition – (H-1)/2 is an integer exactly

when H is odd, mirroring rot180's even-H precondition, and the two are therefore mutually exclusive on one map. It is a true isometry, not an index

permutation: it sends (dq, dr) to (dq+dr, -dr), which permutes the three terms of the axial metric and leaves the distance unchanged. RESERVED BY

DECISION, not blocked on an answer: Q26 is RULED – the enum stays at rot180 | none, so a horizontal mirror is undeclarable and there is nothing

for a gate to verify. Nothing is waiting. It stays reserved rather than deleted because admitting the value later is purely additive – one enum value, this mu, and this invariant – and nothing passing today would then

fail. Contrast T-MOVE-07 above, which IS blocked, on the unruléd Q2.)

T-SCN-11 NON-CONTENTION (Q22 ruled; unit set Q28 ruled): for EACH guidedOpening entry, the OPPOSING seat's cheapest land path to that same `objective` costs STRICTLY MORE MP than the owning seat's lane, as reported by T-SCN-08. The opposing route is minimised over EVERY CanCapture-row unit that seat deploys (Q28), not over that seat's `guidedOpening.infantry` alone: min over the opposing seat's Infantry of cost(hex,

objective)

> the owning lane's cost

PRINT CONVENTION: THE RELATION IS NAMED AT THE SITE, and integer order carries no information – neither which relation is in play nor pass versus fail, since both relations print

the larger integer first on their failing form. Every site says what it is comparing; no site relies on which integer is bigger.

WHAT A BARE PAIR QUANTIFIES OVER, stated because it was only derivable and was read the other way by a careful reader: the right-hand term of a bare "X against Y" is this invariant's opposing term, so it is the MINIMUM over every CanCapture-row unit the opposing seat deploys (Q28) – a set figure, never a cost measured from one named hex. A cost measured from some other named hex is a THIRD quantity; print it with its hex and say what it is – "14 from (1,3) alone, not the set minimum" – so it cannot be mistaken for the set figure. The danger is concrete, not stylistic: the minimising unit can CHANGE under a

counterfactual, so a figure taken from the shipped minimiser stops being the minimum. Asymmetry (ii)'s bullet is exactly that case – excluding the Bridges moves West's minimiser from (1,3) to (1,5), and 14 stopped being the minimum without anything on the map moving.

WHETHER THIS SHOULD HARDEN INTO A CLOSED LIST OF PERMITTED FORMS IS Q30, unruléd and deliberately left so. An earlier revision attempted that codification and withdrew it: each closure outlawed prose elsewhere in the document that was correct, which is a poor trade for a convention that binds wording only. No invariant, fixture, reported integer or refusal condition depends on any of this.

This is the gate for §2.13.1's "uncontested, not merely reachable" – a promise that until this revision was a property of

the drawn map rather than a checkable rule, and one the map failed the first time it was checked (fixture (b)).

IMPLEMENTATION, because the ruled reading looks more expensive than it is: the minimisation is ONE reverse Dijkstra per objective, not one path per unit. Root it at the `objective`

with

$$d(\text{objective}) = 0 \text{ and relax } d(h) = \min \text{ over neighbours } n \text{ of } (\text{MoveCost}(n) + d(n));$$

the result is every hex's cost TO that objective under the T-MOVE-01 accounting, so all of the

opposing seat's Infantry are read off one pass and the cost is independent

of how many units that seat deploys. That identity holds ONLY

because Q21 priced the lane on terrain alone – under an "as deployed" reading each unit sees a different graph and the budget returns to one path per unit. Both sides of the comparison are priced IDENTICALLY, which is what makes the inequality mean anything: Stub-3 cheapest path, terrain alone with occupancy excluded (Q21, ruled), cost counting every hex entered INCLUDING the objective (Factory MoveCost 1) – the T-MOVE-01 accounting T-SCN-06 already uses, so the validator cannot price the two routes two different ways. EQUALITY FAILS. "Strictly longer" is the ruled comparison, and a tie is precisely the race the rule exists to forbid; a >= would have passed the one lane in the PRE-FIX set that a human would call contested – West's South lane against East's second Infantry at (9,5), kept as fixture (b). Q28 moved that Infantry to (9,1), so the set AS SHIPPED holds no such lane today: the failing case survives as a fixture, not as a live refusal, and the operator stays > for the next map that needs it. Three asymmetries with T-SCN-06 are deliberate:

- (i) NO CEILING. T-SCN-06 is a budget ($\leq 2 \times \text{Move}$). This is a comparison: the opposing route may cost anything at all, it must merely cost MORE.
- (ii) BRIDGES ARE ALLOWED on the opposing route, and on the shipped map that allowance is LOAD-BEARING rather than merely permitted. T-SCN-06's Bridge-free clause is a property of the GUIDED lane – what makes the first lesson a walk rather than a crossing – not a constraint the enemy is under. Excluding Bridges would make the opposing route NON-EXISTENT on a bisected map (The Causeway, §2.13.6) and pass that map vacuously, so allowing them is the STRICTER reading and it is the one asserted. Measured PER MAP, never quantified over the set:

- Ferrum Crossing (§2.13.2) EXERCISES it, but on ONE of fixture (a)'s TWO opposing routes, not both. West's cheapest route to North (6,2) – 6 MP from (1,3) – runs over the north Bridge (5,1). TWO Bridge-free figures follow and they are DIFFERENT QUANTITIES, so each is labelled where it stands. FROM (1,3) ALONE, not a set minimum: 14 MP, (2,3)(3,4)(3,5)(4,5)(5,6)(6,6)m(6,5)(6,4)w(6,3)w(6,2) – around the river's southern end, then up through the Woods ring. More than double that hex's own 6. MINIMISED OVER WEST'S INFANTRY – this invariant's opposing term (Q28), and the only figure a BARE pair may carry: 13 MP, from the OTHER West Infantry (1,5), (2,6)(3,6)(3,7)(4,7)(5,7)F(6,7)(7,6)T(6,5)(6,4)w(6,3)w(6,2). Its first five hexes are West's own guided South lane (§2.13.2): 5 MP to (5,7), then 8 MP up the east bank. Bridge-free, West's road to the NORTHERN objective runs through West's own SOUTHERN one. EXCLUDING THE BRIDGES MOVES THE MINIMISER, which is the whole reason the two figures differ. WITH the Bridges, West's set minimum to North is 6, achieved by (1,3); (1,5) alone costs 7, not the set minimum (§2.13.2). WITHOUT them, the set minimum is 13, achieved by (1,5); (1,3) alone costs 14, not the set minimum. THE ACHIEVING UNIT IS NOT THE SAME UNIT under the two readings – that is the flip, and it is why this bullet's two figures are two DIFFERENT quantities rather than one quantity corrected. No "against" is printed in this comparison, deliberately: these are four costs of ONE seat's own units, while an "against" in this stub is the TWO-SEAT inequality a fixture recomputes (PRINT CONVENTION; Q30). The 1 MP is the APPROACH, not the Mountain – (1,5) reaches (5,7) in 5 and (1,3) in 6, while the tail from (5,7) costs 8 for both, and (1,3)'s 14 is achievable Mountain-free as well. So a figure measured from the SHIPPED minimiser is NOT the

counterfactual minimum, and that is the trap this bullet exists to name.

The OTHER opposing route in that fixture is already Bridge-free and does not move: East's cheapest route to South (5,7), 6 MP from (9,3), is (9,4)F(8,5)(8,6)(7,7)(6,7)(5,7), which reaches column 5 only at the objective itself, on row 7, below the river's southern end at (5,5) – the river spans rows 0–5 only. Excluding an edge can only RAISE a shortest path, and that 6 MP witness uses no excluded edge, so 6 stands under either reading. The allowance binds ONE of this map's two objectives, the northern one.

- The Causeway (§2.13.6) EXERCISES it too, passing 3 against 5 in both seats with the crossing permitted; exclude Bridges and it reports "no route" instead of 5 (fixture (c)).

- Longwater March (§2.13.5) is the ONE map with no Bridge on any opposing route, and only because it has no Water and no Bridge hexes at all – its terrain distribution is Water 0, Bridge 0.

WHY THIS IS STATED AS A REASON AND NOT A PERMISSION: a Bridge-free reading does not FAIL Ferrum Crossing in EITHER seat, so no gate in this suite catches it. North still passes, at 5 against 13 – a BARE pair, so its right-hand term is the SET minimum over West's Infantry, the (1,5) route above, and NOT the 14 measured from (1,3) alone. Strictly more than the owning lane on either figure, and either way a number that describes a walk around the entire river. South still passes at 5 against 6, exactly as drawn, because its opposing route never crossed a Bridge. What the counterfactual changes is ONE margin of the two: North's opposing figure goes from 6 to 13 against an unchanged owning 5, widening that margin from 1 MP to 8 MP – a single-digit widening, and an unremarked one, because this invariant asserts a strict inequality and NO CEILING (asymmetry (i)). South's margin stays at 1 MP and never depended on the allowance at all.

THE SPLIT IS THE REASON. The allowance is what keeps the NORTHERN opposing route honest, and neither half of the split is gate-catchable, because no invariant in this stub reads a MARGIN – only the strict inequality, which both seats satisfy either way. So the Bridge is what makes the northern margin a margin, and this suite would go on reporting green while that integer stopped meaning anything.

(iii) THE UNIT SET IS BROADER THAN THE LANE, per Q28

(ruled).

"The opposing seat's cheapest Infantry route" ranges

over

EVERY CanCapture-row unit that seat deploys, not over that seat's own `guidedOpening.infantry` alone. The narrow reading was available, would have passed the shipped map unchanged, and was refused: the property guarded here is a RACE, and a race does not care which Infantry wins it. T-SCN-06's NAMED-hex quantifier is

not

a precedent against this – it names a hex because the guided lane must be the one turn-1a actually marks, whereas nothing is being marked on the opposing side

and

the only question is who can arrive.

deployment

WHAT THE STRICT READING COST, PAID ONCE: one

(9,5),

hex. Ferrum Crossing's East second Infantry was at

with

5 MP from WEST's South objective and therefore tied

(§2.13.2)

West's own 5 MP lane; it now deploys at (9,1)

rule

and the map passes in both seats at 5 against 6. No

as

was weakened, no terrain moved, and the tie survives

part

fixture (b) rather than as an exception.

The relocation was forced, not chosen, which is the

south

worth knowing before anyone edits that map: East's

within

town (7,6) is 2 hexes from South (5,7), so by the triangle inequality any hex covering that town is

5 MP of that factory. On Ferrum Crossing a southern town-capturer IS a racer for West's southern factory – the conflict is geometric, not a placement slip. Reported like T-SCN-08: a refusal carries BOTH measured integers, owning and opposing, so an author reads "5 against 5" rather than "contested" – and a map edit that shortens an enemy approach surfaces as a changed number, not a still-green boolean. Fixtures:

(a) Ferrum Crossing (§2.13.2) PASSES in BOTH seats, reporting 5 against 6 each way: West's South lane 5 from (1,5) against East's cheapest 6 from (9,3); East's North lane 5 from (9,3) against West's cheapest 6 from (1,3) over the north Bridge (5,1). A 1 MP margin each way – the thinnest in the set, on the one map that declares `symmetry: none`. It is also the fixture that catches ASYMMETRIC PRICING: an implementation that counts the objective hex on the owning lane but not on the opposing route reports 5 against 5 and refuses the shipped map.

(b) THE FAILING FIXTURE IS REAL, NOT CONSTRUCTED. The same map with East's second Infantry at its PRE-FIX hex (9,5) – one placement changed, nothing else – must FAIL, reporting 5 against 5: (9,6),(8,7),(7,7),(6,7),(5,7), five cost-1 hexes, and the axial distance from (9,5) to (5,7) is 5, so no cheaper route exists and no implementation detail can make the number anything else. This deployment passed every other invariant in this stub, T-SCN-07's distinctness floor included; T-SCN-11 is the only check in the suite that sees it. It also pins the QUANTIFIER and not merely the comparison: under the Q28 reading REFUSED, (b) passes at 5 against 6, so an implementation that minimises over the guidedOpening unit alone fails this fixture and nothing else in the suite.

(c) The Causeway (§2.13.6) passes 3 against 5 in both seats with the Bridge crossing PERMITTED on the opposing route. It is the fixture for asymmetry (ii): excluding Bridges makes both opposing routes NON-EXISTENT on a bisected map and passes it vacuously, so an implementation that inherits T-SCN-06's Bridge-free clause onto the opposing side reports "no route" here instead of 5.

Determinism: pure parse + validation; any failure refuses the whole file with a reason. scenarioHash is platform-stable by canonical ordering.

The T-SCN-06/08/11 lane costs are Stub-3 path costs and inherit its determinism (T-MOVE-04's canonical tie-break, T-MOVE-06), so the reported integers reproduce across runs and compilers. T-SCN-11 compares two such integers and introduces no new source of nondeterminism.

Acceptance: T-SCN-01..09 and T-SCN-11 headless – the whole written suite. T-SCN-11 ASSERTS from its first run: Q22 gave it the comparison and Q28 gave it the unit set, so this stub carries no written-and-blocked invariant. T-SCN-10 is reserved and UNWRITTEN on Q26, which is a different state: nothing is asserted, so nothing is waiting. T-SCN-11 ships with its three fixtures, one of which – (b), the shipped map's own pre-fix deployment – must FAIL, so the suite demonstrates refusal and not merely agreement with the repo. The §4.2 validate_scenario MCP tool wraps the same checks in-editor for the Content agent; its manual fallback is running the headless validator on the exported file (MCP stays off the critical path, §3 guardrails).

SPEC STUB 8: UI binding contract (Director → UI Scaffolder)

Scope: NOT layout or visual design (§2.11's lane) – this is the contract for how every widget is fed. Widgets bind to a view-model

snapshot plus the §4.9 event list, and hold no rules state (§4.1).

Snapshot fields (read-only, produced by the rules module):

```

per-hex {terrainId, owner}
per-unit {id, side, unitId, hex, hp, hpMax, isFlag, hasActed, captureProgress}
per-side {fameTotal, fameCombat, objectivesHeld X of N, survivingHP}
match {turn, turnCap, sideToMove, resultTier or null}

```

Queries: reachable(unit) → the T-MOVE-01 set; forecast(attacker, defenderHex) → {damage, counterDamage} computed by the verified resolveDamage / defenderCanCounter (5ffa8d6).

Invariants:

- T-UI-01 forecast = resolution: the forecast shown before commit is produced by the same strat call that resolves the attack – identical numbers, mechanically (§2.6, §2.11)
- T-UI-02 the reachable-hex highlight displays exactly the T-MOVE-01 set – the UI queries the module and never recomputes movement (§2.5)
- T-UI-03 the live standings scoreboard (§2.11, §2.8) binds 1:1 to snapshot fields – enemy strength destroyed, objectives held X/N, surviving units/HP, turn vs cap – with no widget-side arithmetic
- T-UI-04 the production menu binds to the buildlist derived from the four Stub-2 unit rows plus current fameTotal; the flag never appears (T-SCN-01's non-producible clause, enforced at the UI layer too)

Determinism: widgets are pure functions of snapshot + events; asserted end-to-end by T-INT-05 (§4.9).

Acceptance: T-UI-01..02 headless (the queries are headless functions); T-UI-03..04 in-editor Automation.

Open questions (Director rulings owed). Every gap found while writing the §4.7 gates (Q1–Q10), the stage-2 additions (Q11–Q13), the rules- and scenario-side rulings folded in here (Q14–Q20), the two gaps found while gating §2.13.1's opening-capture invariant (Q21–Q22), the milestone contradiction the document knowingly carried (Q23), the two raised by the §2.13 symmetry correction (Q24–Q25), the one raised by correcting that correction (Q26), the guided opening's one input-gating constraint (Q27), the reading the Q22 ruling exposed the moment its new invariant was measured against the shipped map (Q28), and the ledger-flip criterion exposed by scoping the week-2 parity gate to its command set (Q29), and the quantifier the T-SCN-11 print convention never stated at a print site, found when a hex-scoped figure was read as a set minimum (Q30) — so that each question carries exactly one ID across the whole document. Each blocks the gate named beside it; the Director writes the rule, the gate then pins it. The last column is not uniform, and the difference matters: **where a reading is stated, it is the conservative one, and it is what ships and what the gates assert** — chosen so that a later ruling loosens behavior rather than invalidating a passing gate. **Q28 was the one row where that convention did not hold**, and how it closed is worth keeping rather than deleting: its conservative reading REFUSED A SHIPPED MAP, so no reading could be stated there without either blocking *Ferrum Crossing* or quietly weakening a rule the Director had just made, and it therefore carried no assumption at all while it was open. It was then ruled the strict way and **the map was corrected instead of the rule** — one deployment hex, no terrain and no rules text (Q28; §2.13.2). The limit of the convention is now known and stated once: **where the conservative reading is not free, this register states no reading and waits. Rows marked *unruled* state no reading and block their gate outright** (Q4's interruption semantics, Q5's stacking, Q6, Q8, and Q9's target- and build-choice ties); those gates and milestones cannot be settled until the Director answers.

ID	Question	Blocks	Assumption in force until ruled
Q1	Map dimensions. §2.2 and §2.10 never state the prototype map's size or shape.	T-HEX-05 bounds; Stub 7's map field	Bounds are per-scenario data, not a global constant; <i>Ferrum Crossing</i> ships 11×9 (§2.13.2).
	Movement classes. §2.3's caption says move cost is	§4.8's MoveClass	§2.3's single cost column applies to every land unit: no

Q2	ID	per movement class and §2.4 gives Questioning “ignores some terrain cost,” but no class set or per-class cost table exists.	column; the reserved T-MOVE-07	Assumption is unruled in place until ruled
Q3		Pass-through. §2.5 pins one-unit-per-hex for <i>ending</i> a move; it is silent on pathing <i>through</i> a friendly-occupied hex.	T-MOVE-03	Blocked — a unit may not path through any occupied hex, friendly or not. N = 1 on the shipped scenario (§2.13.3’s recommendation, inside §2.7’s stated range). Interruption semantics are unruled , so T-FAME-05 stays blocked.
Q4		Capture N and interruption. Pin N (§2.7 says “start N=1–2”), and rule the edge cases: does progress reset if the Infantry leaves or dies mid-capture?	T-FAME-05’s exactness; the §2.9 AI capture step	The §2.7 change request proposes exactly half of each §2.4 cost — 50 / 75 / 100 / 150, all integers. Stacking is unruled .
Q5		Kill-award exact values. “~half its Fame cost” (§2.7) is not assertable. Does the flag’s +500 stack with the Tank’s ordinary kill award?	T-FAME-07	Unruled — unpriced, so no gate asserts it. The rules author recommends (c), with (b) as the fallback; either unblocks the T-CAP- suite immediately.
Q6		Undamaged-strike bonus. “A small bonus” (§2.7) has no number, yet it feeds the tiebreak’s primary key. Options: (a) price it; (b) keep it in the Fame pool but exclude it from the cap tally until priced; (c) cut it — kills already pay half-cost, and the positional triangle already rewards a clean standoff strike with tempo.	T-FAME-07; the T-CAP- tally suite; the kb economy block	Ruled. §2.8, §2.11.4 and §2.13.2 now agree, and T-TURN-04 and Stub 7 read a value rather than an example.
Q7		Turn cap value. RULED (this revision). The cap is per-scenario data , stored in Stub 7’s turnCap; <i>Ferrum Crossing</i> ships 20 turns.	—	
Q8		Income timing and build limits. When within the turn does factory/town income pay, and can it fund a build the same turn? Is there a builds-per-factory-per-turn limit (§2.9’s AI implies one; the player’s rule is unstated)? For a waiting build (§2.7 “the build waits”), is Fame committed at queue time or spawn time, and can it be canceled?	T-FAME-02 and T-FAME-04	Unruled — all three sub-questions. T-FAME-02 and T-FAME-04 stay blocked until ruled, and they gate Stub 4, which §4.11 builds <i>before</i> the turn loop.
Q9		AI tie-breaks. §2.9’s “best expected-damage target” and pathing choices can tie.	T-AI-06; AI determinism as a stated rule rather than an implementation accident	Position ties break by canonical hex order (as gated). Target-choice and build-choice tie order are unruled and need naming so determinism is a rule, not an accident.

ID	Question	Blocks	Assumption in force until ruled
Q10	Flag designation. Confirm exactly one flag per side, designated by the scenario (isFlag, Stub 7), with otherwise-standard Tank stats.	T-SCN-01's exactness, the flag's §4.8 representation	§2.4's "Tank variant" reads as <i>not producible, and nothing else.</i>
Q11	Undo. The command log makes single-step undo of an uncommitted move nearly free, but the GDD never grants it, and §2.6's forecast-then-commit flow arguably forbids it.	Nothing today — no gate assumes undo exists, and §2.11.1's z binding is explicitly conditional	No undo: a completed move stands.
Q12	Zero-RNG confirmation. §2.6 and §4.1 say any RNG "is seeded"; nothing in §2 actually uses RNG.	§4.10's seed field; T-SAVE-01/02 if RNG is ever added	The prototype ships with none , so seed is a reserved field written as 0.
Q13	Player-facing replay. §4.10's format makes a "watch replay" feature cheap, but §2.10 does not scope one.	The §2.10 scope table; a UX handoff if it is scoped	The format stays internal — saves, gates, and balance logs only. It counts for nobody until the objective flips — §2.7's flip-on-capture wording grants nothing before the flip. Partial credit would need a fractional-count rule and would invert T-CAP-05.
Q14	Capture-in-progress at the cap. Does a partially captured objective count toward "objectives held" (§2.8 criterion 2)?	T-CAP-05; the kb victory table	
Q15	The 5-unit standard starting force (§2.13.1) — 1 Flag Tank, 2 Infantry, 1 Artillery, 1 Recon — has no antecedent outside §2.13, alongside map dimensions and town counts.	§2.13.1; Stub 7's deployment list	As drafted in §2.13.1: one of each producible system live from turn 1, a 550-Fame producible force.
Q16	Recon/Air vs. Water. §2.3 marks Water passable by "sea, air"; §2.4's "Recon/Air" never says whether it <i>is</i> air. If Recon crosses Water freely, every bridge chokepoint and <i>The Causeway</i> 's lockout premise leaks.	All three §2.13 maps; the terrain schema	Recon is a land unit with terrain-cost discounts, and bridges bind it. All three maps are priced on this reading.
Q17	Cross-Water Artillery fire. With LOS blocking a stretch goal (§2.2), bank-to-bank fire across a one-hex river is legal at ship, and the two river maps price it per map, not alike : <i>Ferrum Crossing</i> prices contested bank control on it — opposite banks are distance 2, inside Artillery range, so fire crosses before units do (§2.13.2, which states as much explicitly: bridge control there is <i>tempo, not a topological wall</i>); <i>The Causeway</i> prices its Mountain perches on it — range 2–3 covers the bridge hex from +40% cover with no counter (§2.13.6). <i>Longwater March</i> has no	§2.13.2 and §2.13.6 balance, if LOS ever ships	Legal. If LOS blocking ships, Water must not block — or those two maps need a redesign pass.

ID	Question	Blocks	Assumption in force until ruled
Q18	Water and prices nothing on it (§2.13.5) Seat-select scope. Is choosing your seat on the shipped map in scope as “scenario data + one menu affordance”?	§2.13.4’s replay ceiling — 6 configurations if yes, 3 if no	In scope, as §2.13.4 assumes. It is the cheapest replay lever in the document.
Q19	Factory count as a per-scenario dial. <i>Longwater March</i> (§2.13.5) ships 6 factories against §2.7’s “typical ~4 factories total” — inside §2.7’s “two or more neutral” clause, but above its stated typical. Does §2.7’s ~4 describe the shipped map only, leaving count as the match-length dial §2.13.5 uses it as?	§2.13.5; T-SCN-03’s economy check, which currently cites “~4 factories total”	Yes — ~4 describes <i>Ferrum Crossing</i>; §2.13.5’s 6 is a deliberate long-map dial, and T-SCN-03 asserts only the home-plus-two-neutral floor.
Q20	Save/replay milestone split. RULED (this revision). The row is split, not moved: the §4.10 format + headless replayer land in week 2, and only the save-slot UI and slot I/O stay in week 5. Original question: §4.11 showed §4.4’s week-5 save/load placement was one week late in one respect — the format and headless replayer are the instrument for the week-2 integration gate (T-INT-02) and the	§4.4’s milestone table; T-INT-02 and T-SAVE-07 sequencing	Ruled, as §4.11 itself proposed. §4.4’s weeks 2, 4 and 5 and the note under the table now describe one schedule, and the distinction the split turns on is stated there rather than left to be re-derived: a format is a test instrument and ships with the gates that consume it (T-INT-02 first runs wk 2, T-SAVE-07 wk 4); slot I/O is a feature and ships with the rest of the UI. Scheduling-adjacent to Q23 and ruled on the same principle in the opposite direction — a milestone that outran its dependencies moved later; an instrument its own gates outran moved earlier. Amended this revision, and the amendment is the useful part of the row. The ruling was written into §4.4 without re-reading §4.11’s dependency table, which required “rows 1–5 built” for row 9 while Q23 had just limited week 2 to rows 1–3 — so the document promised a gate, and its instrument, in a week its own

ID	Question	Blocks	Assumption in for reproduction
	week-4 self-play logs (T-SAVE-07). Split the row?		That is the Q23 contradiction class one layer down, and the third time these two sections have disagreed. It is repaired by scoping, not by moving: an integration or replay gate is scoped to the command set of the log it runs on, so T-INT-02 runs in week 2 over {Move, Attack} — which already reaches the whole compiler-divergence class it exists to catch, since <code>resolveDamage</code> holds the only non-integer step in the module — and closes in week 3 when rows 4–5 supply the remaining three commands. No week number moved. Whether a partial run may flip a §3 ledger row is registered as Q29 rather than assumed. Ruled as drafted: terrain alone. It reproduces §2.13.1's measured 5/5, 4/4, 3/3 and matches how the lane is actually played, since the other four units move too. Accepted consequence: a map can pass while a seat's own unit sits in the lane on turn 1 — the player walks around it, which on <i>Ferrum Crossing</i>'s 1 MP of slack may cost a turn and is absorbed by beat 2 being a standing directive (§2.11.6-B). Scope, re-checked after the Q28 deployment move rather than assumed. This row's scope was stated as a

ID	Question	Blocks	Assumption in force until ruled and a deployment
Q21	Opening-capture lane measurement: RULED. The lane prices on terrain alone, occupancy excluded. The question was whether T-SCN-06 should price it on the board as deployed — where, under Q3's blocked-pass-through reading, a seat's own four other starting units can make its own lane unmeasurable? The two readings can disagree by several MP on a crowded deployment.	T-SCN-06's pass/fail and T-SCN-08's reported integers; T-SCN-11's four opposing routes on the shipped map — the four cells of §2.13.2's eight-route table that run against a guided lane (East (9,3) and (9,1) → South; West (1,3) and (1,5) → North) — which price on this same convention, as do the table's other four cells: the two guided lanes themselves (T-SCN-06's, not T-SCN-11's) and each seat's second Infantry to its own objective; §2.13.1's three-map lane table if the answer is "as deployed"	property of the drawn deployment, has since moved — East's second Infantry (9,5) → (9,1), §2.13.2 — so it was re-measured. It holds, and at a sharper resolution: no starting unit sits on any of the eight routes Ferrum Crossing now prices — §2.13.2's four-Infantry × two-objective table <i>is</i> the whole priced set, and it already holds the two guided lanes (T-SCN-06) alongside T-SCN-11's four opposing routes and the two same-seat routes that fix each seat's cheaper prize. (9,1) is the origin of two of the eight and lies on the interior of none of them, and both stretch lanes were already clear — so an "as deployed" ruling would move no number on any map as drawn. This row previously read "ten," adding the two guided lanes to a table that already contained them, and called all eight T-SCN-11's; the corrected split is $2 + 4 + 2$. It stays a live question rather than a dead one because of slack — measured per map, not asserted over the set: <i>Ferrum Crossing</i> carries 1 MP against T-SCN-06's ceiling and 1 MP against T-SCN-11's inequality; <i>Longwater March</i> carries 2 MP and 4 MP; <i>The Causeway</i> 3 MP and 2 MP. No figure there is new — each ceiling slack is stated in that man's own section

ID	Question	Blocks	Assumption (if not in force until ruled)
Q22	Uncontested vs. merely reachable. RULED. The validator asserts non-contention: the opposing seat's cheapest Infantry route to the same objective must cost strictly more than the owning seat's lane. T-SCN-07's distinctness clause is a floor beneath that requirement, not the whole of it: two seats can name DIFFERENT objectives and still race each other to one of them, which is exactly what a structural check	T-SCN-07's clause, now a floor; T-SCN-11, written, unblocked and asserting since Q28 ruled; \$4.11 row 7's priced half, which gains	a subtraction of the owning/opposing pair this register already prints at Q22. So <i>Ferrum Crossing</i> is the tightest map in the set on both gates at once, and the one where a deployment edit landing in a priced route is likeliest to flip a gate rather than be absorbed — which is a ranking, not the claim that the other two maps cannot be flipped at all. One further consequence, now that Q22 has widened the surface: terrain-only pricing is what lets T-SCN-11 minimise over a whole seat's Infantry in one reverse Dijkstra per objective (§4.7 Stub 7). An “as deployed” ruling would give every unit its own graph and return that cost to one path per unit. Ruled — and measured against all three maps before the invariant was written, which is what the ruling cost and where it paid. Five of the six lanes cleared as drawn, each printed owning against opposing as T-SCN-11 reports it: <i>Longwater March</i> 4 against 8 in both seats, <i>The Causeway</i> 3 against 5 in both seats, <i>Ferrum Crossing</i>'s East lane 5 against 6. The sixth failed on an exact tie: West's South lane cost 5, and East's second Infantry at (9,5) reached that same objective (5,7) in 5 MP flat

ID	Question	Blocks	Assumption in force until ruled
	cannot see. Gated as T-SCN-11 — T-SCN-10 is reserved-but-unwritten for the horizontal mirror and Q26 keeps it that way, so it was not free to take. Original question: §2.13.1 promises the guided lane is “uncontested, not merely reachable,” but states it as a property of the shipped map rather than a checkable rule.	one full-board pass guidedOpening entry	—Assumption in force until ruled the case the rule exists to catch, it was on the map that ships, and every other invariant in the suite passed it. The map was corrected rather than the rule loosened (Q28): that Infantry now deploys at (9,1), all six lanes pass, and <i>Ferrum Crossing</i> reports 5 against 6 in both seats. The tie is retained as T-SCN-11’s fixture (b) — a failing fixture that was authored rather than constructed.
Q23	Milestone-vs-build-order contradiction. RULED (this revision). §4.11’s critical path makes the baseline AI (row 6) depend on row 5, which depends on row 4 Capture & Fame — but §4.4 promises a working vertical slice <i>with</i> the baseline AI in week 2, and §2.10 states capture + Fame production “land wk 3, not wk 1–2.” The two schedules cannot both hold. Symmetry as a declarable value. RULED. The field stays <code>rot180 \ </code> none with an even-row-count precondition, and <code>rot180</code> on an odd row count is a hard refusal. Original question: §2.13.1 previously offered mirror / rotation / none, but an odd-r offset rectangle has no vertical mirror axis at any dimension, and admits a 180° rotation only when the row count is even. So a vertical mirror is a value the validator could never legally accept, and <code>rot180</code> is only well-formed against an even-H map. It does, however, have a horizontal mirror axis (<code>c ↔ □ c, r ↔ □ H-1-r</code>) exactly when the row count is odd — the geometry the shipped 11 × 9 map sits on (§2.13.1 fact 1, §2.13.2) — so the field is a choice, not a forced hand. Narrow the field to <code>rot180 \ </code> none with an even-row-count precondition, and make <code>rot180</code> on an odd row count a hard refusal rather	§4.4’s week 1–2 milestones; §2.10’s IN row; §4.11’s critical path; and Q20, which is the same decision for save/replay	Ruled: the vertical-slice milestone moves to week 3, and week 2 delivers move + attack only (§4.4). §1, §2.10, §4.4 and §4.11 now describe one schedule — §1’s “core playable” line is the fourth site and was corrected with them.
Q24		§2.13.1’s validation-invariant list; Stub 7’s symmetry field and T-SCN-08’s fixtures (§4.7, tech-director-owned)	Narrowed, as §2.13.1 now reads: <code>rot180 \ </code> none, and <code>rot180</code> declared on an odd row count refuses the file with a reason (a wrong dimension is an authoring error, not a balance question). Both stretch maps are redrawn on 8 rows and declare <code>rot180</code> (§2.13.5, §2.13.6); <i>Ferrum Crossing</i> declares none. If the Director rules otherwise, the only consequence is that a bad declaration surfaces as N failed hex comparisons instead of one refusal — no layout moves.

ID	Question	Blocks	Assumption in force and rule
Q25	than a failed hex comparison? (Whether a third value should be admitted for the horizontal mirror is Q26, which owns that question — this row asks only about the narrowing.)		What should a horizontal-mirror value be answered at Q26, not here.
	What a rot180 declaration binds. RULED. Terrain + ownership (sides exchanged) + placements (sides exchanged); <code>guidedOpening</code> is not bound. Original question: §2.13.1 says declared symmetry is machine-verified but never says over <i>what</i>. Terrain only? Terrain + ownership + placements — the reading both stretch maps are actually drawn to (§2.13.5 “every one of those is a ρ-pair”; §2.13.6 “East is the exact ρ-image of West”)? And does it bind <code>guidedOpening</code>, so the two seats’ lanes must themselves be ρ-images?	T-SCN-09’s assertion set; and T-SCN-08’s fixture (b), whose equal 4 / 4 is a <i>theorem</i> if <code>guidedOpening</code> is bound and only a <i>measurement</i> if it is not	Terrain + ownership (sides exchanged) + placements (sides exchanged), as gated in T-SCN-09. <code>guidedOpening</code> is not bound: §2.13.1 requires each seat’s lane to be its own neutral within 6 MP, never that the two be images of each other, so a rot180 map may legitimately name non-image lanes and report a split. Costless either way today — both stretch maps satisfy the <i>stronger</i> reading as drawn (<i>Longwater</i>: $\rho(1,2) = (11,5)$ and $\rho(4,1) = (8,6)$; <i>Causeway</i>: $\rho(1,2) = (7,5)$ and $\rho(3,2) = (5,5)$), so ruling <code>guidedOpening</code> in would fail no shipped map, and ruling terrain-only would merely loosen T-SCN-09. The enum stays <code>rot180 \ none</code> and T-SCN-10 stays unwritten: adding a value to a REQUIRED enum moves every scenario’s hash, and no shipped map needs it — <i>Ferrum Crossing</i> declares none because its terrain is asymmetric, and both stretch maps are even-H <code>rot180</code>. The conservative reading is the narrow one, so a later ruling only ever <i>widens</i> the accepted set and <i>adds</i> an assertion; nothing that passes today would start failing. The cost of leaving it narrow, stated plainly: an
	Is a horizontal mirror declarable? RULED — no, ruled together with Q24 as one decision. The enum stays at two values and T-SCN-10 stays unwritten; a horizontal mirror may be added later, purely additively, if a map ever needs one. Original question: Q24 narrowed the enum to <code>rot180 \ none</code> partly on the ground that mirror is a value the validator could never legally accept. That holds for the vertical axis at every dimension, but not for the horizontal one: $\mu(c, r) = (c \text{ } H - 1 - r)$ is a genuine	Stub 7’s symmetry field (§4.7); the reserved T-SCN-10; nothing else — no gate asserts	

Q26 ID	Question	Blocks	Assumption in force until ruled
	isometry of an odd-r rectangle whenever the count is odd — H-1 is then even, so r and H-1-r share a parity, the offset is preserved and the column is unchanged. <i>Ferrum Crossing</i> is 11×9 and carries that axis geometrically (§2.13.1). So Q24's enum survives, but as a scope decision, not an impossibility. Add <code>mirrorH</code> with an odd-row-count precondition — the exact counterpart of <code>rot180</code>'s even-row-count one — or keep the enum at two values and accept that a true horizontal mirror is undeclarable?	anything about a value the schema does not admit, and no §2.13 map is drawn to a horizontal mirror	author who genuinely draws a horizontal mirror on an odd-H map must declare none, and none asserts nothing — a verifiable property is silently discarded rather than an authoring error caught, which is the opposite of the failure mode Q24 was choosing between. Note the two are never in competition: <code>rot180</code> needs even H, a horizontal mirror needs odd H, and their composition would be the vertical mirror that never exists, so at most one non-none value is well-formed on any given map.
Q27	Is gating End Turn during beat 1a presentation or rule? RULED. It ships as specified: End Turn is inert during beat 1a until the marked Infantry has moved. Teaching by constraint is accepted here because it guarantees beat 1a retires inside turn 1, which is what makes §2.11.6-B's schedule predictable. Original question: During beat 1a of a guided opening only, End Turn is inert until the marked Infantry (<code>guidedOpening.infantry</code>) has moved; hover reads Move the marked Infantry first. It is scoped to the first match, dies with <code>Skip</code> guidance, and never applies outside the guided window — but it is the only guided-opening constraint that gates a player <i>input</i> rather than a selection, which is why it is registered rather than assumed.	Nothing today, and nothing in §4.7: no stub or T- ID gates the directive strip, and Stub 7 deliberately keeps the guidance layer out of Stub 8's snapshot. The dependency is internal to §2 — §2.11.6-B's turn-1 row is unconditional in all three branches only if 1a cannot outlive turn 1, so a ruling of “no input gating” would require that row to be re-derived	Ruled: ships as specified. The alternative, recorded because it was weighed: a player who ends turn 1 without moving leaves 1a outstanding, and rule 1 hands the strip to 1b (End turn.) — an instruction they have just followed — so the fallback is to let 1a expire silently at the turn boundary like 1b, and §2.11.6-B's turn-1 row gains a footnote. Ruled the strict way, knowingly and at a stated price. (b) was available and would have passed the shipped map untouched; it was refused because the property Q22 protects is a race, and a race does not care which <code>Infantrv</code> wins it. T-

ID	Question	Blocks	Assumption and force unit ruled
Q28	Whose Infantry the T-SCN-11 opposing route is measured from: RULED (this revision). Reading (a): the opposing route is minimised over any Infantry that seat owns, not over its <code>guidedOpening.infantry</code> alone. Original question: Q22 ruled that the opposing seat's cheapest Infantry route to a guided objective must cost strictly more than the owning seat's lane, but not over which units "cheapest" ranges — (a) every <code>CanCapture-row</code> unit that seat deploys, since either Infantry can race, or (b) that seat's own <code>guidedOpening.infantry</code> alone, which keeps the comparison lane-against-lane and matches how T-SCN-06 quantifies over a NAMED hex rather than an existential. Ledger-flip criterion for a partially-scoped gate. §4.11 rows 9–10 now run their gates in week 2 over a	Nothing further. T-SCN-11 is unblocked and asserting (§4.7 Stub 7), and §4.11 row 7 carries it in its acceptance set	Assumption and force unit ruled: not a counter-precedent — it names a hex because the guided lane must be the one turn-1a marks, whereas nothing is marked on the opposing side and the only question is who can arrive. The cost was one deployment move, not a weakened rule. East's second Infantry moved (9,5) → (9,1) (§2.13.2); no terrain, factory or town count, lane cost, home-factory-empty rule or turn estimate moved with it, and the relocation was forced rather than chosen — East's south town (7,6) is 2 hexes from South (5,7), so by the triangle inequality any hex covering that town races that factory, and the only free southern hexes clearing 5 MP were the Artillery's and the Flag Tank's. <i>Ferrum Crossing</i> now reports 5 against 6 in both seats — 1 MP each way, the thinnest margin in the set. The prefix 5 against 5 is kept as T-SCN-11's fixture (b): a failing case that was actually authored, that passes every other invariant in the suite, and that reading (b) would have passed. Conservative reading in force: a row flips only when its full acceptance set passes over the complete §4.9 command set at one commit; a partial pass is reported as a run and never as a closure, and §4.11

ID	Question	Blocks	Assumption in force until ruled
Q29	{Move, Attack} log and re-run them on the complete command set in week 3 (Q20, amended). §3's ledger says a row is verified when it cites "the commit and passing test IDs that back it," and §3's two bars require agent-authored tests that pass plus a human sign-off — but neither says whether <i>passing</i> means the acceptance set ran over the system's whole input domain. Without a rule, a green week-2 T-INT-02 could flip a proposed ledger row while the log it replays is missing three of the five §4.9 commands.	§3's ledger rows for §4.9 and §4.10 (both proposed rows, both unwritten today); no gate — every test runs either way, this governs only what may be <i>claimed</i> from a run	rows 9-10 are written in exactly those two parts. Free in the conservative direction — it can only delay a claim, so a later loosening invalidates nothing that passed. The alternative worth weighing, since the information is real and currently discarded: record partial passes in the ledger as a dated "<i>green over subset X at commit Y</i>" line, which reports more without claiming more — that is a §3 presentation decision, not a technical one, which is why it is registered rather than assumed. Partial reading in force, deliberately not hardened. Two things are settled and written into the convention: a bare pair is set-quantified (the right-hand term is the minimum over every CanCapture unit the opposing seat deploys, per Q28), and a cost measured from some other named hex is printed with its hex and what it is — "14 from (1,3) alone, not the set minimum". Naming a hex beside a bare pair stays legal when that hex is the minimiser, which is what fixture (a) does. The counterfactual figures are corrected to this reading: 5 against 13, margin 8 MP, with 14 retained as the explicitly (1,3)-scoped figure it always was. What is left open is whether this should harden into a closed list of permitted forms.
Q30	What a T-SCN-11 "against" print quantifies over, and what form a hex-scoped route cost takes. A withdrawn revision of the print convention (§4.7 Stub 7) named two relations for an "against" and gave two printed forms — a bare pair for owning-against-opposing, a named ceiling for measured-against-budget — but there are three quantities in play, and when this row was filed the third had neither a printed form nor an exclusion. The bare pair's right-hand term is this invariant's opposing term and is therefore the minimum over every CanCapture-row unit the opposing seat deploys (Q28); that is derivable from the formula, was never stated at a print site, and was read the other way by a reader looking specifically for it. A cost measured from one named hex is neither of the two printed relations, so it has nowhere to go but the bare form, which means something else. That is mechanically how asymmetry (ii)'s Bridge-free counterfactual came to print "5 against 14": 14 is West's Bridge-free cost from (1,3)	Nothing computable. T-SCN-11's inputs, formula, unit set, reported integers, refusal conditions and all three fixtures are identical either way, and no map, lane cost or deployment moves. It governs only what §4.7's prose may claim from a measured integer — which is where the error	

ID	Question (1,5)	Blocks	Assumption in force until ruled
	alone, while the set minimum — (2,6)(3,6)(3,7)(4,7)(5,7)F(6,7)(7,6)T(6,5)(6,4)w(6,3)w(6,2), West’s own guided South lane plus 8 MP up the east bank. The two differ because excluding the Bridges moves the minimising unit : with them permitted West’s set minimum is 6, achieved by (1,3), and (1,5) alone costs 7; without them the set minimum is 13, achieved by (1,5), and (1,3) alone costs 14. Should the convention gain a third printed form for a hex-scoped cost (“14 from (1,3) alone, not the set minimum”), or should a hex-scoped cost be forbidden inside an “against” print entirely?	was, and is the one place this suite has no gate.	Assumption in force until ruled and it was withdrawn: each version outlawed prose elsewhere in the document that was correct — a within-seat ranking in §2.13.2, an objective-count comparison in §2.13.5 — and twice broke its own rule inside the passage stating it. That is a poor trade for a convention binding wording only, so the question is left to a human rather than half-answered. Nothing computable turns on it: no invariant, fixture, reported integer or refusal condition depends on any of it.

4.8 Data contract — DataTable schemas

Principle: authored once, read twice, proven equal. Each table is one canonical CSV in the repo (data/). The headless loader parses it directly; the Unreal editor imports the same file into a UDataTable whose row struct derives FTableRowBase. A parity gate (T-DATA-05, Unreal Automation) iterates every imported row and asserts it equals the CSV field-for-field. Nothing is authored twice, so the headless sim and the engine can never disagree about a stat. Missing column or unparseable value = hard load failure, never a silent default.

Unit schema — data/units.csv → headless strat::UnitDef → UStruct FUnitRow. Exactly four rows (Infantry, Tank, Artillery, Recon; §2.4). The **flag unit is not a row**: §2.4 defines it as “a designated Tank,” so flag status is a placement-level field in the scenario file (isFlag, §4.7 Stub 7), gated by T-SCN-01 — not a fifth unit type. One representation, one gate.

Column	CSV type	Headless field (strat::UnitDef)	UStruct field (FUnitRow)	Source
Id (row name)	string	id	RowName (FName)	§2.4
HP	int	hpMax	int32 HP	§2.4 (10/20/8/12)
Move	int	move	int32 Move	§2.4 (3/5/3/7)
Atk	int	atk	int32 Atk	§2.4 (4/8/10/5)
Def	int	def	int32 Def	§2.4 (2/5/1/3)
RangeMin	int	rangeMin	int32 RangeMin	§2.4 (Artillery 2, others 1)
RangeMax	int	rangeMax	int32 RangeMax	§2.4 (Artillery 3, others 1)
CostFame	int	costFame	int32 CostFame	§2.4 (100/300/200/150)
Type	enum string	strat::UnitType type	EUnitType Type	addendum Part A — order fixed: Infantry, Tank, Artillery, Recon
CanCapture	bool	canCapture	bool bCanCapture	§2.7 (Infantry only)
MoveClass	enum string	<i>reserved</i>	<i>reserved</i>	blocked on Q2

EUnitType is a UENUM mirroring strat::UnitType (Combat.h, addendum Part A) with the enumerator order pinned; T-DATA-05 asserts the mirror is exact so an editor-side reorder

can never silently reindex the effectiveness table below.

Terrain schema — `data/terrain.csv` → `headless strat::TerrainDef` → `UStruct FTerrainRow`. Exactly seven rows (§2.3).

Column	CSV type	Headless field	UStruct field
Id (row name)	string	id	RowName
MoveCost	int (0 = impassable)	moveCost	int32 MoveCost
DefensePct	int, signed	defensePct	int32 DefensePct
PassLand/PassAir/PassSea	bool ×3	passLand/Air/Sea	bool bPassLand/Air/Sea
Capturable	bool	capturable	bool bCapturable
IncomeFame	int	incomeFame	int32 IncomeFame
IsSpawnPoint	bool	isSpawnPoint	bool bIsSpawnPoint
IsRepairPoint	bool	isRepairPoint	bool bIsRepairPoint

Type-effectiveness schema — `data/effectiveness.csv` → `strat::effectiveness` → `UStruct FEffectivenessRow`. A 4×4 matrix, row = attacker type, columns = defender types, values ∈ {0.5, 1.0, 1.5} (§3 spec), **shipping all-1.0** (§2.4 — the triangle stays positional). The verified implementation (`Combat.cpp::effectiveness @ 5ffa8d6`) hardcodes the neutral stub; this schema is the lever’s *data* form, so that if self-play ever asks for a non-1.0 cell, populating it is a CSV edit gated by the existing directional-gate plan (addendum Part A), not a code change. T-COMBAT-09 (neutral stub, 16/16 pairs) continues to pin the shipped state; a non-neutral CSV with T-COMBAT-09 still in the suite is a deliberate, visible gate change the Director must approve — the “do not invent balance values” rule enforced by the pipeline itself.

Invariants (the T-DATA set – Stub 2, §4.7):

T-DATA-01 loaded unit values equal the §2.4 table exactly (4 rows, all columns)

T-DATA-02 loaded terrain values equal the §2.3 table exactly (7 rows), including Bridge’s NEGATIVE defense and Water impassable-to-land

T-DATA-03 exactly one unit row has `CanCapture == true` (Infantry, §2.7)

T-DATA-04 sanity: all costs > 0; `RangeMin` ≤ `RangeMax`; `HP` > 0

T-DATA-05 (editor, Unreal Automation) every imported `DataTable` row equals the

CSV field-for-field, and `EUnitType` mirrors `strat::UnitType` exactly

T-DATA-06 `effectiveness.csv` is 4×4, indexed in the pinned type order, every

cell ∈ {0.5, 1.0, 1.5}; the SHIPPED file is all-1.0 (re-asserting

T-COMBAT-09 at the data layer)

Determinism: pure parse; missing/malformed field = hard fail, no defaults.

Acceptance: T-DATA-01..04, 06 headless; T-DATA-05 in the editor pass.

4.9 Headless-module → Unreal integration path

The rules module’s value is that it has **zero engine dependencies** (§3, §4.1); integration

must add Unreal *around* it without ever adding Unreal *to* it.

1. Module layout — one source, two compilers. The certified headless sources (Combat.h/Combat.cpp today; each §4.7 stub as it lands) live canonically in the crew repo, where the g++/clang++ + python run.py gate runs (§3 ledger). The UE project vendors them verbatim into a UBT runtime module, Source/StratRules/, via a sync script that records the source commit hash. StratRules contains **no engine headers, no UObject, no third-party includes** — pure C++17 in namespace strat, exactly the base-spec constraint. The standalone gate keeps compiling the identical files, so “the engine build works” never substitutes for “the gate passed.”

2. Bridge — the only code that knows both worlds. The game module (Stratocracy) owns: - **Load:** FUnitRow/FTerrainRow/FEffectivenessRow → strat::UnitDef / strat::TerrainDef / effectiveness table (a mechanical §4.8 mapping), plus strat::loadScenario on the shipped scenario asset (§4.7 Stub 7). - **The authoritative strat::GameState.** Actors and UMG hold no rules state (§4.1 “never own rules” — here made structural, not aspirational). - **Command in / events out.** Presentation submits commands — Move{unit, destHex}, Attack{unit, targetHex}, Build{factoryHex, unitId}, Capture{unit}, EndTurn{} — the rules module validates then applies each, and emits an **ordered, deterministic event list:** Moved(path), Damaged, Destroyed, Captured, Spawned, BuildWaiting, Repaired, IncomePaid, MatchEnded(tier). Actors and widgets animate and rebind **from events and the view-model snapshot only** (§4.7 Stub 8). An invalid command returns a rejection reason and changes nothing. - **Threading:** synchronous on the game thread. A full turn resolution is microseconds of integer math; the shipping AI “moves instantly” (§2.8). No async, and no MCP involvement anywhere in the runtime path — the MCP plugin remains editor-only tooling, experimental, off the critical path (§3 guardrails, §4.2, §4.5).

3. Parity gates. The command log format of §4.10 is the instrument: the same recorded match is replayed by the headless harness and by an in-editor Automation test through the UBT-compiled module, and both must land on the same canonical state hash. This is what catches the one genuinely engine-shaped risk — a compiler/CRT divergence in the damage formula’s round — mechanically instead of by playtest anecdote.

```
SPEC STUB: Integration parity                (Director → Systems Engineer /
UI Scaffolder)
Inputs:  vendored StratRules sources + recorded source commit; a §4.10
replay          file; the §4.8 tables imported in-editor.
Invariants:
  T-INT-01  source identity: every file in Source/StratRules/ hash-matches
the          recorded crew commit — the ledger's evidence chain survives
vending
  T-INT-02  replay parity: the same command log replayed headless and in-
engine      (Automation test) produces the same final canonical state
hash.      (The tripwire of this stub: an agent that "ports" rather than
vendors    the module — or a compiler that rounds differently — passes
every      behavior test in one world and silently diverges in the
other.)
  T-INT-03  rejection safety: an illegal command leaves the state hash
unchanged  and returns a reason; no partial application
  T-INT-04  no engine deps: StratRules compiles standalone under the
existing    g++/clang++ gate — the gate run itself is the assert
  T-INT-05  (editor, Automation) presentation statelessness: after any
event      sequence, rebuilding all widgets/actors from the current view-
model      snapshot alone reproduces the same displayed values (nothing
lives      only in a widget)
Determinism: the bridge never reorders, drops, or synthesizes events.
Acceptance: T-INT-01, 04 on every gate run; T-INT-02, 03, 05 in the editor
pass.
```

4.10 Save & replay format

Design choice: a save *is* a replay. Because every system is a deterministic pure function or state machine (§2.6, §4.1, and every §4.7 determinism gate), the cheapest correct save is not a state snapshot but **the scenario reference plus the ordered command log**. Loading = loadScenario + re-apply the log — headless-speed, sub-second. One format,

four consumers:

1. **Single-slot save/load** (§4.1’s “minimal single-slot,” now specified).
2. **The T-INT-02 parity gate** — its input file is a save file.
3. **Balance Analyst self-play logs** (§4.1 harness) — each self-play match is emitted as this same format, so every balance claim in the ledger’s evidence chain is a *replayable* artifact, not a summary statistic.
4. **Bug reproduction** — a playtest failure attaches its save; any agent can replay it headless.

File layout — versioned JSON, one file:

Field	Type	Meaning
formatVersion	int	This layout’s version; unknown = refuse load
rulesCommit	string	Crew commit of the rules module that wrote the file (T-INT-01’s hash)
dataHash	string	Hash of the §4.8 CSV set in effect
scenarioId / scenarioHash	string	The §4.7 Stub-7 scenario file and its hash
seed	int	Reserved; written as 0 — no RNG ships (§2.6, pending Q12)
commandLog	array	The ordered commands of §4.9, exactly as the bridge consumed them, tagged {turn, side}
stateHash	string	Canonical hash of the resulting state (integrity check)
result	string/null	Result tier (§2.8) if the match ended, else null

Canonical state hash. Defined once, in the headless module: serialize the GameState in a fixed field order — turn counter, side to move, per-side fameTotal/fameCombat, objective ownership, per-unit {id, side, hex, hp, isFlag, captureProgress, pendingBuilds} sorted by the canonical hex order (§4.7 conventions, T-HEX-07) — then hash the bytes. Every field is an **integer** (eff and the HP ratio exist only transiently inside resolveDamage), so the hash is platform-stable by construction; T-INT-02 proves it across compilers.

Policies (prototype): - **Single slot** — one file (slot0), stored via a thin USaveGame wrapper holding the JSON so platform save paths are respected. Overwrite-confirm UX is **unowned**: §2.11 specifies no save/load surface, and neither its screen list (§2.11.5) nor its build ranking (§2.11.8) includes one — either §2.11 gains the surface, or the prototype ships silent single-slot overwrite. - **Save points** — a save is accepted between atomic commands during the player’s turn. The AI turn resolves synchronously in one call (§4.9), so “mid-AI-turn” is not a reachable save state by construction. - **Mid-match saves** — the log up to the last completed command *is* the save. Derived pending state (a waiting build, capture-in-progress) replays correctly because it is a function of the log. - **Version policy** — mismatched formatVersion, rulesCommit, dataHash, or scenarioHash → **refuse load with a reason**. No migration in the prototype; a save is only valid against the exact rules and data that wrote it.

SPEC STUB: Save & replay (Director → Systems Engineer)
Inputs: scenario file (Stub 7), command log, §4.8 data set, module commit.
Invariants:
T-SAVE-01 round-trip: play N commands → save → load → identical stateHash
T-SAVE-02 replay determinism: the same file loaded twice → identical hashes
(leans on every §4.7 determinism gate – T-HEX-07, T-MOVE-06, T-FAME-09, T-TURN-09, T-AI-06; this is their end-to-end composition)
T-SAVE-03 prefix validity: every prefix of a valid log is itself a valid,
loadable save (mid-match save falls out of this)
T-SAVE-04 refusal: any header mismatch (version/rules/data/scenario hash) →
load refused with a reason; state untouched
T-SAVE-05 no partial load: a log with an illegal command at index k is refused whole; the pre-load state survives. (The tripwire: an agent
state that applies-then-validates leaves a corrupted half-loaded state
that passes every happy-path test.)
T-SAVE-06 stateHash stability: hash of a given state is identical across the
headless and in-engine builds (asserted jointly with T-INT-02)
T-SAVE-07 harness compatibility: a Balance Analyst self-play log validates
and replays as a save file – one format, no dialect drift
Determinism: pure; the file contains everything needed to reproduce the state.
Acceptance: T-SAVE-01..07 headless (05 also exercised in-editor via the load UI
path); slot I/O smoke test in the editor pass.

4.11 Build order

Rows 1–8 are the §4.7 stubs (the eight **pending** ledger rows); rows 9–10 are the §4.9 and §4.10 systems. Combat, its test suite, Repair, and Type-effectiveness are green at 5ffa8d6 and are prerequisites, not work items. Note **row 10's format spec must exist by the row-9 integration pass**, because T-INT-02's input file is a §4.10 save. Rows 9 and 10 therefore state their dependencies in **two parts** — what each gate needs in order to **run**, and what it needs in order to **close** — because both are scoped to the command set of the log they replay, and week 2's log carries only {Move, Attack} (§4.4). Reading either row's requirement as a single set is what put a gate in a week this table could not supply it.

#	System (ledger row)	Depends on	Headless?	Acceptance test IDs
1	Hex grid & math (§4.7 Stub 1)	— (Q1 pins bounds)	Yes	T-HEX-01..07
2	Data tables (§4.8, incl. effectiveness CSV)	— (MoveClass column blocked on Q2)	Loader + T-DATA-06 yes; import parity in-editor	T-DATA-01..06
3	Movement & pathfinding (Stub 3)	1, 2	Yes	T-MOVE-01..06
4	Capture & Fame economy (Stub 4)	3	Yes	T-FAME-01..09
5	Turn loop & win/tiebreak (Stub 5)	4 + verified Combat/Repair @ 5ffa8d6	Yes	T-TURN-01..09
6	Opponent AI (Stub 6)	5	Yes	T-AI-01..06 + self-play smoke
		1, 2 for the structural half (T-SCN-01..03, 05, 07, 09); 3 for the priced half — T-SCN-04, 06, 08, 11 all cost a path, and T-SCN-11 costs a full-board pass rather than a path : it minimises over every Infantry the opposing cost	Yes; MCP tool	T-SCN-01..09, 11

7 #	Scenario File & validator (Stub 7)	Infantry the opposing seat deploys (Q28, ruled), which is one reverse	wraps it in-editor; manual fallback stands	(10 reserved test IDs unwritten on Q26)
		Dijkstra per guidedOpening objective and is therefore independent of that seat's unit count — done naively as one path per opposing unit it scales with the deployment instead		
8	UI binding (Stub 8)	5, 7 (snapshot needs full state)	Contract + queries yes; widgets in-editor	T-UI-01..04
9	Presentation bridge & integration — §4.9 (proposed ledger row)	Run vs close. Vendoring and T-INT-01/04 depend on no rules row at all — the sync script and the standalone gate run <i>are</i> the assert — and close as soon as vendoring lands. T-INT-02/03/05 need only the rows behind the log they replay: rows 1–3 for week 2's {Move, Attack} log, and they re-open when rows 4–5 add Capture/Build/EndTurn. They close on rows 1–5 , which is what this cell used to state as the whole dependency	Source/compile gates yes; replay parity + statelessness in-editor	T-INT-01..05
10	Save & replay — §4.10 (proposed ledger row)	Three parts, three dependency sets. (a) <i>Format spec</i> + <i>header/version machinery</i> — no deps at all; write it first , and T-SAVE-04 (refusal on any header mismatch) closes on it alone, since it never applies a command. (b) <i>Headless replay</i> — rows 1–3, plus row 7's structural half for the scenarioId/scenarioHash it loads; it runs T-SAVE-01/02/03/05/06 over week 2's {Move, Attack} log. (c) <i>Closure</i> — rows 4, 5 complete the command set (T-SAVE-01/03/05/06), row 6 completes T-SAVE-02's determinism composition, and T-SAVE-07 needs row 6's self-play besides. Slot I/O is week 5 and no headless gate waits on it	Yes, all but slot I/O	T-SAVE-01..07

Critical path: 1 → 3 → 4 → 5 → 6/8. Row 2 runs in parallel immediately. **Row 7 no longer sits beside the chain; it straddles it.** Its structural half (T-SCN-01..03, 05, 07, 09) starts once 1–2 land, but **four of its ten written invariants** — T-SCN-04's flag reachability and T-SCN-06/08/11's opening-capture lane — price a Stub-3 path, so row 7 cannot *close* until row 3 does. T-SCN-11 (Q22 and Q28, both ruled) is the newest of the four and the most expensive: it prices the *opposing* seat's cheapest route to the same objective, minimised over every Infantry that seat deploys, which is one reverse Dijkstra per *guidedOpening* objective — a full-board pass rather than a path, and one whose cost does not grow with the deployment. It **asserts**, as of this revision: nothing in row 7 is written-and-blocked, and the only unwritten invariant is T-SCN-10, reserved on Q26. It ships with a **failing** fixture — the shipped map's own pre-fix deployment, which tied 5

against 5 — so this row’s suite proves it can refuse a real scenario and not merely agree with the repo. Row 7 is still not ON the critical path (nothing in the chain waits on it), but scheduling it as “parallel from week 1” would leave its ledger row un-flippable and the §2.11.6 guided opening ungated for however long movement slips: the scenario row flips after movement, not before. 6 and 8 fork after 5. **Row 10 is split, per Q20 (ruled).** Its *format and headless replayer* are not week-5 content but the instrument two earlier gates run on — T-INT-02’s input file is a save, and the week-4 self-play logs T-SAVE-07 validates are the same format — so §4.4 now lands them beside the scenario loading they already sit next to, and leaves only the **save-slot UI and slot I/O** at the end. That is the split this section proposed and the Director adopted; §4.4’s table and the note under it are the single statement of the schedule, and this paragraph deliberately does not restate the week numbers a second time.